PROBABILITY & STATISTICS

A First-Principles Guide for Complete Beginners

Expanded from CSE 301 Lecture Notes (Lectures 1–17)

Original Notes by Tahmid Hasan

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Contents

1. Introduction to Probability

1.1 What Is This Subject About?

Probability is a branch of mathematics that gives us precise tools for reasoning about uncertainty. Before any formulas appear, it helps to understand the motivation: in the real world we constantly face situations where the outcome is not known in advance — will it rain tomorrow, will a coin land heads, will a patient who tests positive actually have a disease. Probability theory does not pretend to eliminate this uncertainty; instead, it gives us a rigorous language to measure it, manipulate it, and draw correct conclusions from it. A line attributed to the founders of modern statistics captures this idea well: *statistics is the logic of uncertainty*. Just as ordinary logic gives rules for reasoning when facts are certain (if A implies B, and A is true, then B is true), probability gives rules for reasoning when facts are uncertain.

1.1.1 A Note on History

The mathematical study of probability has a long history, and a few names recur throughout these notes:

- **Fermat and Pascal** (17th century) are usually credited as founders of probability theory, originally motivated by questions about fair division of stakes in gambling games that were interrupted early.
- **Newton** also engaged with probability questions, including the Newton–Pepys problem discussed later in these notes.
- The **Mosteller–Wallace authorship problem** is a famous case where statistical methods (analyzing word-frequency patterns) were used to settle a historical dispute about who actually wrote certain anonymous essays (the *Federalist Papers*). This illustrates that probability and statistics are not just abstract mathematics — they are tools for answering real factual questions when direct evidence is unavailable.

1.2 Sample Spaces and Events

To reason about an uncertain situation mathematically, we must first describe precisely what outcomes are possible. This is the role of two foundational definitions.

Definition. Sample Space

The **sample space** S of an experiment is the set of *all possible outcomes* of that experiment.

The word "experiment" here is used in a very general sense: it just means any process whose result is not predetermined — flipping a coin, rolling a die, measuring a patient's temperature, or even something complicated like an election. The sample space must list every outcome that could conceivably happen, and the outcomes are normally chosen so that exactly one of them occurs.

Definition. Event

An **event** is a subset of the sample space, i.e., $A \subseteq S$. We say the event A *occurs* if the actual outcome of the experiment is a member of A.

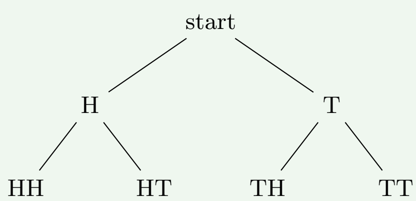
For example, if the experiment is rolling a six-sided die, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$. The statement "the die shows an even number" corresponds to the event $A = \{2, 4, 6\}$, which is indeed a subset of S. If the die actually lands on 4, then since $4 \in A$, the event A has occurred.

1.2.1 Building a Sample Space with a Tree Diagram

When an experiment consists of several stages, it is often useful to build the sample space using a **tree diagram**: a branching picture where each path from the root to a leaf represents one possible outcome of the whole experiment.

Worked Example. Flipping a coin twice

Consider flipping a fair coin two times in a row. The first flip can be Heads (H) or Tails (T). For *each* of those two possibilities, the second flip can again be H or T. We can draw this branching structure as a tree:



Reading off the four leaves of the tree, the sample space for two coin flips is

$$S = \{HH, HT, TH, TT\}.$$

Each leaf is reached by a unique path (first flip, then second flip), and the tree diagram makes it visually clear that there are exactly $2 \times 2 = 4$ outcomes: two choices at the first branching, multiplied by two choices at the second branching for each of those. This multiplying-of-branches idea is the seed of the **multiplication rule**, which we formalize in Section 1.4.

1.3 The Naive Definition of Probability

Once we have a sample space, the simplest possible way to assign probabilities is to assume that every individual outcome is equally likely, and then say the probability of an event is the fraction of outcomes belonging to that event.

Definition. Naive Definition of Probability

If a sample space S has n outcomes, all of which are equally likely, and an event A consists of k of those outcomes, then

$$\mathbb{P}(A) = \frac{\#\text{favorable outcomes}}{\#\text{possible outcomes}} = \frac{k}{n}.$$

The crucial hidden assumption here, stated explicitly in the box's name, is that *all outcomes in S are equally likely*. This is not always true! For a fair coin or fair die, the assumption is reasonable by symmetry. But if, say, a die is weighted so that 6 comes up more often than the others, the naive definition simply does not apply — we would need a more general framework (introduced in Section 2.3) that allows different outcomes to have different probabilities.

In the coin-flipping example above, all four outcomes HH, HT, TH, TT are equally likely (assuming a fair, unbiased coin and independent flips), each with probability $1/4$. The event "at least one head" is $\{HH, HT, TH\}$, which has 3 outcomes out of 4, so its naive probability is $3/4$.

1.4 Counting: The Multiplication Rule

Many probability problems boil down to counting: how many outcomes are in the sample space, and how many are in the event of interest? Because writing out every single outcome (as we did with the tree above) becomes impossible once the numbers get large, we need systematic counting rules. The most basic and important one is the **multiplication rule**.

Suppose an experiment consists of r sub-experiments performed in sequence. If the first sub-experiment has n_1 possible outcomes, and *for each outcome of the first*, the second sub-experiment has n_2 possible outcomes, and so on, so that for each combination of outcomes of the first $r-1$ sub-experiments the r-th sub-experiment has n_r possible outcomes, then the total number of possible outcomes of the overall experiment is

$$n_1 \times n_2 \times \cdots \times n_r.$$

Why is this true? Picture the tree diagram again. At the first level there are n_1 branches. From *each* of those n_1 branches, n_2 new branches sprout, giving $n_1 n_2$ branches after two levels. Continuing this way for r levels, the number of leaves — which is the number of complete outcome sequences — is the product $n_1 n_2 \cdots n_r$. This matches exactly what we saw with two coin flips: $n_1 = 2$ (first flip), $n_2 = 2$ (second flip), giving $2 \times 2 = 4$ total outcomes.

1.4.1 Worked Example: Probability of a Full House in Poker

This example uses the multiplication rule together with a counting tool called the **binomial coefficient**, written $\binom{n}{k}$, which counts the number of ways to choose k objects out of n when the order of selection does not matter (so choosing cards A, B is the same as choosing B, A). We will derive and explain $\binom{n}{k}$ properly in Chapter 2; for now, it suffices to know that $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$ and that, for instance, $\binom{4}{2} = 6$ counts the number of ways to pick 2 items out of 4.

Worked Example. Full house in a 5-card poker hand

A standard deck has 52 cards: 13 ranks (Ace, 2, 3, ..., 10, Jack, Queen, King) and, for each rank, 4 suits (clubs, diamonds, hearts, spades). A **full house** is a 5-card hand consisting of exactly three cards of one rank (called "the triple") and exactly two cards of a different rank (called "the pair"). We want $\mathbb{P}(\text{full house})$ when 5 cards are dealt uniformly at random from the deck (so the naive definition applies, with every possible 5-card hand equally likely).

Step 1 — count the total number of 5-card hands. Since order doesn't matter when dealing a hand, the total number of possible hands is $\binom{52}{5}$. This is our denominator: the total number of equally likely outcomes.

Step 2 — count the number of ways to choose the rank used for the triple, and the suits within that rank. There are 13 ranks to choose from for the triple. Once the rank is fixed (say we chose "Kings"), we must choose which 3 of the 4 suits appear in our hand; there are $\binom{4}{3} = 4$ ways to do this. By the multiplication rule, the number of ways to pick the rank-and-suits for the triple is

$$13 \times \binom{4}{3} = 13 \times 4.$$

Step 3 — count the number of ways to choose the rank for the pair, and the suits within that rank. The pair must use a *different* rank from the triple, so there are only 12 ranks left to choose from. Once that rank is fixed, we choose 2 of its 4 suits, in $\binom{4}{2} = 6$ ways. So the number of ways to pick the rank-and-suits for the pair is

$$12 \times \binom{4}{2} = 12 \times 6.$$

Step 4 — combine via the multiplication rule. The choice of triple (rank + suits) and the choice of pair (rank + suits) are independent sub-choices performed in sequence, so by the multiplication rule the total number of full-house hands is the product of the two counts above:

$$13\binom{4}{3} \cdot 12\binom{4}{2}.$$

Step 5 — divide by the total number of hands. Applying the naive definition of probability,

$$\mathbb{P}(\text{full house}) = \frac{13\binom{4}{3} \cdot 12\binom{4}{2}}{\binom{52}{5}}.$$

Plugging in numbers: $\binom{4}{3} = 4$, $\binom{4}{2} = 6$, and $\binom{52}{5} = 2{,}598{,}960$. The numerator is $13 \times 4 \times 12 \times 6 = 3744$. So

$$\mathbb{P}(\text{full house}) = \frac{3744}{2{,}598{,}960} \approx 0.00144,$$

or roughly 0.144% — about 1 in 694 hands. This matches the well-known rarity of full houses in poker, and illustrates the general strategy used throughout this subject: identify the sample space, count the favorable outcomes using the multiplication rule (breaking the choice into independent stages), and divide.

2. Counting and Story Proofs

2.1 The Sampling Table

A huge fraction of counting problems can be reduced to one question: *in how many ways can we choose k objects out of n objects?* The answer depends on two binary choices we must make about the rules of selection:

- Does **order matter**? (Is choosing A then B different from choosing B then A?)
- Is selection **with replacement** or **without replacement**? (Can the same object be chosen more than once?)

These two choices give four combinations, summarized in the following table, called the **sampling table**.

	Order matters	**Order doesn't**
With repl.	n^k	$\binom{n+k-1}{k}$
Without repl.	$\dfrac{n!}{(n-k)!}$	$\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$

Let us understand each cell from first principles.

With replacement, order matters (n^k). Here we are choosing k objects one at a time, putting each one back (or equivalently, the same object can be picked again) before the next choice, and we care about the sequence in which they were chosen. By the multiplication rule from Chapter 1, there are n choices for the first pick, n choices for the second pick (since replacement means the pool is unchanged), and so on for k picks, giving $n \times n \times \cdots \times n = n^k$.

Without replacement, order matters ($n!/(n-k)!$). Now each object can be used at most once. There are n choices for the first pick, but only $n-1$ remaining choices for the second pick (since one object is now used up), then $n-2$ for the third, and so on, down to $n-k+1$ choices for the k-th pick. By the multiplication rule, the count is $n(n-1)(n-2)\cdots(n-k+1)$, which can be written compactly as $n!/(n-k)!$ because $n! = n(n-1)\cdots(n-k+1)\cdot(n-k)!$, and dividing by $(n-k)!$ cancels the tail of the factorial.

Without replacement, order doesn't matter ($\binom{n}{k}$). This is the most commonly used count, and is called "n choose k." We can derive it from the previous cell: there are $n!/(n-k)!$ ways to choose k objects *in order*, but if order does not matter, every *unordered* group of k objects has been counted multiple times — once for each of the $k!$ orders in which that same group could have been selected. So we must divide by $k!$ to remove this over-counting:

$$\binom{n}{k} = \frac{n!/(n-k)!}{k!} = \frac{n!}{k!(n-k)!}.$$

With replacement, order doesn't matter ($\binom{n+k-1}{k}$). This is the trickiest cell, and deserves its own explanation, given next.

2.2 Stars and Bars

Suppose we want to choose k objects from n types, with replacement allowed (the same type can be chosen multiple times), and we don't care about the order in which we made the choices — we only care about *how many of each type* we ended up with. This is exactly equivalent to a different-looking problem:

Equivalent reformulation

Choosing k indistinguishable objects with replacement from n types (order irrelevant) is the same as asking: *how many ways are there to place k indistinguishable balls into n distinguishable boxes?* (Box i represents "how many times type i was chosen.")

To count this, picture the k balls as k identical dots ("stars") arranged in a row, and use $n-1$ dividers ("bars") to split the row of stars into n groups — the dots before the first bar go into box 1, the dots between the first and second bar go into box 2, and so on, with the dots after the last bar going into box n.

box 1		box 2		box 3
$\star\ \star$	$\vert$	$\star$	$\vert$	$\star\ \star$

This picture is just one example with $k = 4$ stars and $n = 3$ boxes (so $n - 1 = 2$ bars). The key realization is that any arrangement of k stars and $n-1$ bars in a row corresponds to exactly one way of distributing k balls into n boxes, and vice versa. So the question becomes: how many ways are there to arrange a row of $k + (n-1)$ symbols, where k of them are stars and $n-1$ of them are bars? This is the same as choosing which k of the $k+n-1$ positions in the row will be stars (the rest are automatically bars), which is, by definition,

$$\binom{n+k-1}{k}.$$

This confirms the fourth cell of the sampling table.

2.3 Story Proofs

A **story proof** is a proof technique where, instead of manipulating algebraic formulas, we show that two expressions are equal by describing a single counting scenario (a "story") and showing that both sides of the equation count the very same thing, just in two different ways. This is often more illuminating than algebra because it explains *why* an identity is true, not just that it is true.

2.3.1 Story Proof 1: Symmetry of the Binomial Coefficient

$$\binom{n}{k} = \binom{n}{n-k}$$

Story: Imagine choosing a committee of k people from a group of n people. The left side, $\binom{n}{k}$, directly counts the number of ways to do this. But choosing *which k people are on the committee* is exactly equivalent to choosing *which $n-k$ people are left out* (not on the committee) — once you know who is excluded, you automatically know who is included, and vice versa. So the number of ways to choose who is *not* on the committee, $\binom{n}{n-k}$, must count exactly the same set of outcomes. Hence the two expressions are equal.

2.3.2 Story Proof 2: The Captain Identity

$$n\binom{n-1}{k-1} = k\binom{n}{k}$$

Story: Consider the task of choosing a committee of k people out of n, and then designating one member of that committee as "captain." We count this in two ways.

Way 1 (right-hand side): First choose the committee of k people, which can be done in $\binom{n}{k}$ ways. Then, for each such committee, choose which of its k members is the captain, which can be done in k ways. By the multiplication rule, the total number of (committee, captain) pairs is $k\binom{n}{k}$.

Way 2 (left-hand side): First choose who the captain is, directly out of all n people — there are n ways to do this. Then choose the remaining $k-1$ committee members from the $n-1$ people who are not the captain, which can be done in $\binom{n-1}{k-1}$ ways. By the multiplication rule, the total is $n\binom{n-1}{k-1}$.

Both ways count exactly the same set of (committee, captain) pairs, just by deciding the captain at a different point in the process. Therefore the two counts must be equal: $n\binom{n-1}{k-1} = k\binom{n}{k}$.

2.3.3 Story Proof 3: Vandermonde's Identity

$$\binom{m+n}{k} = \sum_{j=0}^{k}\binom{m}{j}\binom{n}{k-j}$$

Story: Suppose we have a group of m men and n women, $m+n$ people total, and we want to choose a committee of k people from this combined group. The left side, $\binom{m+n}{k}$, directly counts the number of such committees.

For the right side, split the selection by how many of the k committee members are men. If exactly j of the k members are men (j can range from 0 to k), then we choose those j men from the m available men in $\binom{m}{j}$ ways, and we choose the remaining $k-j$ members (who must be women) from the n available women in $\binom{n}{k-j}$ ways. By the multiplication rule, for a fixed value of j, there are $\binom{m}{j}\binom{n}{k-j}$ committees with exactly j men. Since these cases (for different values of j) are mutually exclusive and cover every possibility, we sum over all j from 0 to k to get the total count:

$$\sum_{j=0}^{k}\binom{m}{j}\binom{n}{k-j}.$$

Since both sides count the same committees, they are equal. This identity will resurface later (Chapter 8) in proving that the sum of two independent Binomial random variables is again Binomial.

2.4 The Axiomatic (Non-Naive) Definition of Probability

The naive definition of probability required all outcomes to be equally likely, which is too restrictive for most real situations (a biased coin, a weighted die, or any experiment where outcomes have inherently different likelihoods). We now give a fully general definition that does not require this assumption.

Definition. Probability Space

A **probability space** consists of a sample space S together with a function $\mathbb{P}$ that takes any event $A \subseteq S$ as input and returns a number $\mathbb{P}(A) \in [0, 1]$, subject to the following two axioms:

1. $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(S) = 1$. (The impossible event has probability 0; the event "something happens" — the entire sample space — has probability 1.)
2. For any countable collection of events $A_1, A_2, A_3, \ldots$ that are **disjoint** (no two of them share an outcome, i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$),
$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

This second axiom is called **countable additivity**. Intuitively, it says that if events cannot happen simultaneously (they're disjoint), then the probability that *at least one of them* happens is simply the sum of their individual probabilities — there's no overlap to worry about, so no double-counting and no need for any correction term. This axiomatic framework, due originally to the mathematician Andrey Kolmogorov, is the foundation underlying everything else in this subject; the naive definition from Chapter 1 is just a special case of it, valid when every individual outcome is assigned the same probability $1/n$.

It's worth pausing on why these two simple axioms are enough to build an entire theory. Every other property of probability that we will use — complements, unions of overlapping events, conditional probability, and so on — can be *derived* from just these two rules, as we will see in the next chapter. This is analogous to how a small set of axioms in geometry (Euclid's postulates) can be used to logically derive every theorem in geometry.

3. The Birthday Problem and Properties of Probability

3.1 The Birthday Problem

The **birthday problem** asks: given k people in a room, what is the probability that at least two of them share a birthday (same day of the year, ignoring leap years so there are 365 possible birthdays)?

The trivial case. If $k > 365$, then by the **pigeonhole principle** (if you have more pigeons than pigeonholes, at least two pigeons must share a hole), it is *certain* that two people share a birthday, since there are only 365 possible birthdays but more than 365 people. So $\mathbb{P}(\text{match}) = 1$ whenever $k > 365$.

The interesting case, $k \leq 365$. We assume that each person's birthday is equally likely to be any of the 365 days, and that everyone's birthday is independent of everyone else's (knowing one person's birthday tells you nothing about another's). Directly computing $\mathbb{P}(\text{at least one match})$ is hard, because there are many different ways a match could occur (persons 1 and 2 match, or 1 and 3, or 2 and 5, etc., or multiple matches at once). Instead, we use a standard trick: compute the probability of the *complementary* event, "no match at all," which is much easier, and then subtract from 1.

Computing $\mathbb{P}$(no match). Imagine assigning birthdays to the k people one at a time. The first person's birthday can be anything: 365 choices, no constraint yet. For there to be no match, the second person's birthday must differ from the first person's, so there are 364 acceptable choices out of 365 possible. The third person's birthday must differ from *both* of the first two, giving 363 acceptable choices, and so on, until the k-th person, who has $365 - k + 1 = 366 - k$ acceptable choices. By the multiplication rule, the number of ways to assign birthdays with no match is $365 \cdot 364 \cdots (365 - k + 1)$. The total number of ways to assign birthdays with no restriction is 365^k (each of the k people independently has 365 choices). So by the naive definition,

$$\mathbb{P}(\text{no match}) = \frac{365 \cdot 364 \cdots (365 - k + 1)}{365^k}.$$

And therefore

$$\mathbb{P}(\text{match}) = 1 - \mathbb{P}(\text{no match}) = 1 - \frac{365 \cdot 364 \cdots (365 - k + 1)}{365^k}.$$

The numbers that come out of this formula are famously counterintuitive:

k	$\mathbb{P}(\text{match})$
23	$\approx 50.7\%$
50	$\approx 97\%$
100	$\approx 99.9994\%$

The surprise is that with only 23 people, there is already better than a coin-flip's chance that two of them share a birthday, even though there are 365 days to "spread out" over. The intuitive mistake people make is thinking about whether *their own* birthday matches someone else's (which is unlikely, since it's 1 specific day among 365), rather than realizing that with 23 people there are $\binom{23}{2} = 253$ *pairs* of people, and each pair has some chance of matching — with that many pairs being checked, a match becomes quite likely.

3.2 Basic Properties of Probability

Recall from Chapter 2 that the axiomatic definition of probability rests on just two axioms. We now derive several useful consequences — properties that hold for *any* valid probability function, not just naive/equally-likely settings.

3.2.1 Property 1: Complement Rule

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$$

Here A^c (read "A complement") denotes the event that A does *not* occur, i.e., $A^c = S \setminus A$, the set of all outcomes not in A.

Proof. Note that A and A^c are disjoint (an outcome cannot both be in A and not in A), and together they make up the entire sample space: $A \cup A^c = S$. By axiom 2 (additivity for disjoint events),

$$\mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c).$$

But $\mathbb{P}(A \cup A^c) = \mathbb{P}(S) = 1$ by axiom 1. So $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$, which rearranges to $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. ■

This is an extremely useful trick (as we already saw implicitly in the birthday problem): when an event is complicated, but its complement is simple, compute the complement's probability and subtract from 1.

3.2.2 Property 2: Monotonicity

If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Proof. Since $A \subseteq B$, every outcome of A is also an outcome of B. We can write B as a disjoint union: $B = A \cup (B \cap A^c)$, where $B \cap A^c$ is the part of B that is *not* in A. (These two pieces are disjoint, and together they reconstruct exactly B.) By axiom 2,

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c).$$

Since probabilities are always ≥ 0 (this follows from $\mathbb{P}(A) \in [0, 1]$ in the axioms), $\mathbb{P}(B \cap A^c) \geq 0$, so $\mathbb{P}(B) \geq \mathbb{P}(A)$. ■
In words: a bigger event (a superset) can never be less likely than a smaller event (a subset) it contains.

3.2.3 Property 3: Inclusion–Exclusion for Two Events

When two events A and B can overlap, we cannot simply add $\mathbb{P}(A) + \mathbb{P}(B)$ to get $\mathbb{P}(A \cup B)$, because outcomes in the overlap $A \cap B$ would be counted twice. The fix is the **inclusion–exclusion** formula:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Proof. Decompose $A \cup B$ into two disjoint pieces: A itself, and the part of B not already in A, i.e., $B \cap A^c$. So

$$\mathbb{P}(A \cup B) = \mathbb{P}(A \cup (B \cap A^c)) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c)$$

using axiom 2, since A and $B \cap A^c$ are disjoint. It remains to show that $\mathbb{P}(B \cap A^c) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$. To see this, note that B itself splits into two disjoint pieces: the part of B that overlaps with A (namely $A \cap B$), and the part of B that does not overlap with A (namely $B \cap A^c$). So

$$\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(B \cap A^c),$$

which rearranges to $\mathbb{P}(B \cap A^c) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$. Substituting this back into our first equation gives

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \qquad \blacksquare$$

The subtracted term $\mathbb{P}(A \cap B)$ corrects for the double-counting of outcomes that belong to both A and B.

3.2.4 Property 4: Inclusion–Exclusion for Three Events

Extending this idea to three events requires more correction terms, since now there are overlaps between every pair, plus a triple overlap counted incorrectly by the pairwise corrections.

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) &+ \mathbb{P}(B) + \mathbb{P}(C) \\ &- \mathbb{P}(A \cap B) - \mathbb{P}(B \cap C) - \mathbb{P}(C \cap A) \\ &+ \mathbb{P}(A \cap B \cap C)\end{aligned}$$

The reasoning generalizes the two-event case: start by adding all three individual probabilities, but any outcome lying in exactly two of the three sets has now been counted twice, so we subtract each pairwise intersection once. However, an outcome lying in *all three* sets was counted three times in the first sum, then subtracted three times in the pairwise corrections (once for each pair), netting to zero — so we must add back $\mathbb{P}(A \cap B \cap C)$ once to correctly count it exactly once overall.

3.2.5 Property 5: General Inclusion–Exclusion

The pattern continues for any number of events n:

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_i \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i A_j) + \cdots$$

where the sum continues, alternating in sign, through all single events, all pairs, all triples, and so on up to the single n-fold intersection $A_1 \cap \cdots \cap A_n$.

This formula is used extensively in the next section to solve the matching problem.

3.3 De Montmort's Matching Problem (1708)

This classical problem, due to the French mathematician Pierre Rémond de Montmort, sets up a scenario where inclusion–exclusion is essential (no shortcut works).

Worked Example. Setup

Shuffle a deck of n cards labeled $1, 2, \ldots, n$ uniformly at random (every ordering is equally likely). Define A_i to be the event that card i ends up in position i (a "match" at position i). We want $\mathbb{P}$(at least one match), i.e., $\mathbb{P}(A_1 \cup A_2 \cup \cdots \cup A_n)$.

We address this fully in Chapter 4, where we compute each piece of the inclusion–exclusion sum and find a beautiful limiting answer.

4. The Matching Problem and Independence

4.1 Solving the Matching Problem

Continuing from Chapter 3, we now compute each ingredient needed for the inclusion–exclusion formula applied to the matching problem.

Single event probability, $\mathbb{P}(A_i)$. The event A_i says card i lands in position i. By symmetry, every position is equally likely to hold card i under a uniformly random shuffle, so the probability that it lands specifically in position i is $1/n$:

$$\mathbb{P}(A_i) = \frac{1}{n}.$$

(More formally: there are $n!$ total shuffles, and $(n-1)!$ of them have card i fixed at position i, since the remaining $n-1$ cards can be arranged freely; so $\mathbb{P}(A_i) = (n-1)!/n! = 1/n$.)

Pairwise intersection, $\mathbb{P}(A_i \cap A_j)$. This is the probability that *both* card i is in position i and card j is in position j simultaneously. Once these two cards' positions are fixed, the remaining $n-2$ cards can be arranged in any of $(n-2)!$ ways, out of $n!$ total shuffles:

$$\mathbb{P}(A_i \cap A_j) = \frac{(n-2)!}{n!}.$$

General k-fold intersection. By the same logic, fixing k specific cards to their matching positions leaves $n-k$ cards free to permute in $(n-k)!$ ways:

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \frac{(n-k)!}{n!}.$$

Applying inclusion–exclusion. We now plug these into the general formula. The sum $\sum_i \mathbb{P}(A_i)$ has $\binom{n}{1} = n$ terms, each equal to $1/n$, but instead of summing identical terms one by one we use the binomial coefficient to count how many terms appear at each level: there are $\binom{n}{k}$ ways to choose which k indices appear in a k-fold intersection, and each such intersection has probability $(n-k)!/n!$. So

$$\mathbb{P}\left(\bigcup_i A_i\right) = \binom{n}{1} \cdot \frac{1}{n} - \binom{n}{2} \cdot \frac{1}{n(n-1)} + \binom{n}{3} \cdot \frac{(n-3)!}{n!} - \cdots$$

Let's simplify each term. The k-th term is $\binom{n}{k} \cdot \frac{(n-k)!}{n!}$. Using $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, this becomes

$$\binom{n}{k} \cdot \frac{(n-k)!}{n!} = \frac{n!}{k!(n-k)!} \cdot \frac{(n-k)!}{n!} = \frac{1}{k!}.$$

Every term collapses beautifully to just $1/k!$. So the whole sum is

$$\mathbb{P}\left(\bigcup_i A_i\right) = \frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} - \cdots \pm \frac{1}{n!} = 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots \pm \frac{1}{n!}.$$

(The first term $1/1! = 1$ is rewritten as is; conventionally we start the displayed sum from the $1/2!$ term being subtracted.)

4.1.1 The Limiting Value as $n \to \infty$

What happens to this probability as the number of cards n grows very large? Recall the Taylor series for e^{-1}:

$$e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots$$

Our sum $1 - \frac{1}{2!} + \frac{1}{3!} - \cdots$ is exactly $1 - e^{-1}$ in the limit (matching term by term against the series above, after noticing the sign pattern lines up once we account for the leading 1). Therefore, as $n \to \infty$,

$$\mathbb{P}(\text{at least one match}) \to 1 - e^{-1} \approx 0.632.$$

Why this is surprising

No matter how large the deck is — 52 cards, 1000 cards, a million cards — the probability that *at least one* card ends up in its original position converges to a fixed constant near 63.2%, rather than going to 0 or 1. Intuitively, even though each individual card has a vanishingly small chance ($1/n$) of matching as n grows, there are also n cards each getting that small chance, and these two effects (smaller probability per card, but more cards) balance out to leave a non-trivial constant probability.

Aside. The lecture notes mention, as a side remark unrelated to probability, **Strassen's algorithm** for fast matrix multiplication, which multiplies two $n \times n$ matrices in roughly $O(n^{2.81})$ time rather than the naive $O(n^3)$, by cleverly recombining sub-block products. This is purely a computer-science aside and has no further bearing on the probability content.

4.2 Independence of Events

So far we've talked about events without asking whether knowing one tells us anything about another. The notion of **independence** formalizes "knowing about one event gives no information about another."

Definition. Independence of Two Events

Events A and B are **independent** if
$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

Why is this the right definition? Intuitively, if A and B are unrelated, then the chance both happen should just be the product of their individual chances — this is exactly how probabilities of unrelated coin flips multiply (e.g., $\mathbb{P}(\text{first flip H and second flip H}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, matching what we'd get from the naive definition on the sample space $\{HH, HT, TH, TT\}$). We will see in Chapter 5 that this definition is equivalent to saying $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ — i.e., learning that B occurred does not change the probability we assign to A.

Definition. Independence of Three Events

Events A, B, C are independent if they are **pairwise independent** — meaning
$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B), \quad \mathbb{P}(B \cap C) = \mathbb{P}(B)\mathbb{P}(C),$$
$$\mathbb{P}(C \cap A) = \mathbb{P}(C)\mathbb{P}(A)$$
— *and additionally*
$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\,\mathbb{P}(B)\,\mathbb{P}(C).$$

It's important to notice that pairwise independence *alone* is not enough to guarantee the full three-way condition; both the three pairwise conditions and the triple-product condition must hold simultaneously. (There exist constructed examples where three events are pairwise independent but the triple intersection condition fails, showing the extra condition really is necessary and not automatic.)

4.3 The Newton–Pepys Problem (1693)

This problem was posed to Isaac Newton by Samuel Pepys, and it's a good illustration of independence combined with the complement trick.

Worked Example. Which is more likely?

Compare the probabilities of three events when rolling fair six-sided dice:

- A: at least one 6 appears in 6 dice rolls.
- B: at least two 6s appear in 12 dice rolls.
- C: at least three 6s appear in 18 dice rolls.

Computing $\mathbb{P}(A)$. Use the complement trick: it's easier to find $\mathbb{P}(\text{no 6s at all in 6 rolls})$. Each individual die independently shows a non-6 with probability $5/6$, and since the rolls are independent, the probability all 6 rolls avoid a 6 is $(5/6)^6$ (multiplying the independent probabilities, by the very definition of independence extended to many events). So
$$\mathbb{P}(A) = 1 - \left(\frac{5}{6}\right)^6 \approx 0.665.$$

Computing $\mathbb{P}(B)$. Here we want at least *two* 6s among 12 rolls, so the complement ("fewer than two 6s," i.e., zero or exactly one 6) is slightly more involved. The probability of zero 6s in 12 rolls is $(5/6)^{12}$ by the same reasoning as before. The probability of *exactly one* 6 in 12 rolls requires choosing which of the 12 rolls is the 6 (there are 12 ways to choose this position), with that roll showing 6 (probability $1/6$) and the other 11 rolls all avoiding 6 (probability $(5/6)^{11}$); so this probability is $12 \cdot \frac{1}{6}\left(\frac{5}{6}\right)^{11}$. (This expression is a preview of the Binomial PMF formalized in Chapter 8.) Putting these together,
$$\mathbb{P}(B) = 1 - \left(\frac{5}{6}\right)^{12} - 12 \cdot \frac{1}{6}\left(\frac{5}{6}\right)^{11} \approx 0.619.$$

Computing $\mathbb{P}(C)$. Similarly, for at least three 6s in 18 rolls, the complement is "zero, one, or two 6s," so we subtract the probabilities of exactly k sixes for $k = 0, 1, 2$:
$$\mathbb{P}(C) = 1 - \sum_{k=0}^{2} \binom{18}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{18-k} \approx 0.597.$$

Conclusion. Putting the three numbers side by side,
$$\mathbb{P}(A) \approx 0.665 > \mathbb{P}(B) \approx 0.619 > \mathbb{P}(C) \approx 0.597,$$
so event A (at least one 6 in 6 rolls) is the most likely of the three, even though all three scenarios are built by "scaling up" the number of dice and the required number of 6s in the same proportion (6:1, 12:2, 18:3). This shows that scaling up a probability problem proportionally does not preserve the probability — a subtlety that is easy to get wrong by intuition alone, and is exactly why working the algebra out carefully matters.

4.4 Introducing Conditional Probability

All of the examples above raise a natural question: how should we update our probabilities once we learn that some event has occurred? This is the subject of **conditional probability**, formally defined as

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Conditional probability is how we mathematically represent updating our belief about event A in light of new evidence that event B has occurred. We explore this central concept in full detail in the next chapter.

5. Conditional Probability

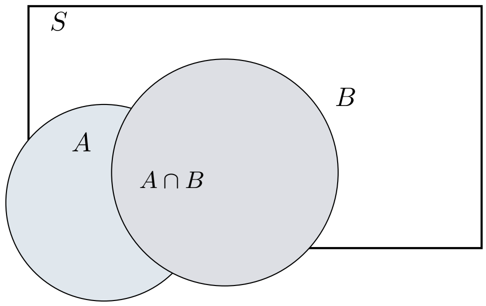
5.1 Definition and Intuition

Definition. Conditional Probability

For events A and B with $\mathbb{P}(B) > 0$, the **conditional probability of A given B** is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The intuition: once we know that B has occurred, the sample space effectively shrinks down to just B — any outcome outside B is now known to be impossible. Within this new, smaller universe B, the event A can only happen via the outcomes that are in *both* A and B, i.e., $A \cap B$. So we re-normalize: we take the probability of $A \cap B$ (within the original space) and divide by the probability of the new universe B, so that probabilities within this new universe still sum to 1. The picture below illustrates this.



If we restrict attention to the region B (shaded), the "fraction of B" that is also covered by A is precisely the overlapping region $A \cap B$ relative to all of B — which is exactly the ratio $\mathbb{P}(A \cap B)/\mathbb{P}(B)$.

5.2 Theorem 1: The Two Ways to Write a Joint Probability

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B) = \mathbb{P}(A)\,\mathbb{P}(B \mid A)$$

This follows immediately by multiplying both sides of the definition of conditional probability by $\mathbb{P}(B)$ (giving $\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B)$), and symmetrically by swapping the roles of A and B (giving $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)$, using $\mathbb{P}(B \mid A) = \mathbb{P}(A \cap B)/\mathbb{P}(A)$). Since $A \cap B$ is the same set regardless of which letter we call "first," both expressions must be equal. This innocent-looking rearrangement is actually one of the most-used tools in the whole subject, because it lets us compute a joint probability $\mathbb{P}(A \cap B)$ by breaking it into a sequential story: first the probability that B happens, times the probability that A happens *given* that B already happened.

5.3 Theorem 2: The Chain Rule

The idea above generalizes to any number of events, peeling them off one at a time.

$$\mathbb{P}(A_1, \dots, A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1) \cdots \\ \cdots \mathbb{P}(A_n \mid A_1, \dots, A_{n-1})$$

(Here $\mathbb{P}(A_1, A_2, \dots, A_n)$ is shorthand for $\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n)$, the probability that all the events happen together.) The chain rule is proven by repeatedly applying Theorem 1: first split off A_n to get $\mathbb{P}(A_1, \dots, A_n) = \mathbb{P}(A_1, \dots, A_{n-1})\,\mathbb{P}(A_n \mid A_1, \dots, A_{n-1})$, then split off A_{n-1} from the remaining joint probability the same way, and so on, until only $\mathbb{P}(A_1)$ remains un-split. This rule is especially useful for sequential experiments, such as drawing cards one at a time without replacement: the probability of a specific sequence of draws is the product of each draw's probability *conditional on* all the previous draws.

5.4 Theorem 3: Bayes' Rule

Bayes' Rule is perhaps the single most important formula in this entire subject, because it tells us how to *reverse* a conditional probability — that is, how to get from $\mathbb{P}(B \mid A)$ to $\mathbb{P}(A \mid B)$, which is often what we actually want to know.

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)}$$

Derivation. From Theorem 1, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)$. Plugging this into the definition of $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ immediately gives Bayes' Rule. The power of this formula is that sometimes $\mathbb{P}(B \mid A)$ is easy to know or measure directly (for instance, the known accuracy of a medical test, $\mathbb{P}(\text{positive test} \mid \text{disease})$), while $\mathbb{P}(A \mid B)$ — the thing we actually care about (e.g., $\mathbb{P}(\text{disease} \mid \text{positive test})$) — is hard to measure directly but can be *computed* from the other quantities using Bayes' Rule.

5.5 The Law of Total Probability

To use Bayes' Rule in practice, we usually need to compute the denominator $\mathbb{P}(B)$, which itself often requires breaking B down according to different possible "causes" or scenarios.

If $A_1, A_2, \ldots, A_n$ partition the sample space (they are disjoint and their union is everything, i.e., exactly one of them must occur), then for any event B,

$$\begin{aligned}\mathbb{P}(B) = \mathbb{P}(A_1)\mathbb{P}(B \mid A_1) + \mathbb{P}(A_2)\mathbb{P}(B \mid A_2)\\ + \cdots + \mathbb{P}(A_n)\mathbb{P}(B \mid A_n).\end{aligned}$$

Why this is true. Since $A_1, \ldots, A_n$ partition the sample space, the event B can be split into disjoint pieces according to which A_i occurred alongside it: $B = (B \cap A_1) \cup (B \cap A_2) \cup \cdots \cup (B \cap A_n)$, and these pieces are disjoint (since the A_i are disjoint). By axiom 2 of probability (additivity over disjoint events),

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B \cap A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i)\mathbb{P}(B \mid A_i),$$

where the last step uses Theorem 1 on each term. This is exactly the Law of Total Probability. Intuitively: to find the overall probability of B, break the world into mutually exclusive scenarios A_i, find how likely B is *within* each scenario, weight by how likely each scenario is, and add up.

5.6 Worked Example: Aces in a Two-Card Hand

Worked Example. Comparing two conditional probabilities

Deal a random 2-card hand from a standard 52-card deck (so all $\binom{52}{2}$ hands are equally likely). We compare two different conditional probabilities:

$$\mathbb{P}(\text{both aces} \mid \text{have an ace})$$

versus

$$\mathbb{P}(\text{both aces} \mid \text{have the ace of spades}).$$

These sound similar but are not the same question — one conditions on having *some* ace (any of the 4), the other on having a *specific* ace (the ace of spades).

Computing $\mathbb{P}$(both aces). This is needed for both calculations. There are $\binom{4}{2} = 6$ ways to choose 2 aces out of the 4 aces in the deck, out of $\binom{52}{2}$ total 2-card hands, so

$$\mathbb{P}(\text{both aces}) = \frac{\binom{4}{2}}{\binom{52}{2}}.$$

Computing $\mathbb{P}$(have an ace). It's easier to use the complement: $\mathbb{P}(\text{have an ace}) = 1 - \mathbb{P}(\text{no ace at all})$. The number of hands with no aces is $\binom{48}{2}$ (choosing both cards from the 48 non-ace cards), so

$$\mathbb{P}(\text{have an ace}) = 1 - \frac{\binom{48}{2}}{\binom{52}{2}}.$$

First conditional probability.

$$\mathbb{P}(\text{both aces} \mid \text{have an ace}) = \frac{\mathbb{P}(\text{both aces})}{\mathbb{P}(\text{have an ace})} = \frac{\binom{4}{2}/\binom{52}{2}}{1 - \binom{48}{2}/\binom{52}{2}}.$$

Computing the pieces: $\binom{4}{2} = 6$, $\binom{48}{2} = 1128$, $\binom{52}{2} = 1326$. So the numerator is $6/1326$ and the denominator is $1 - 1128/1326 = 198/1326$. Dividing,

$$\mathbb{P}(\text{both aces} \mid \text{have an ace}) = \frac{6}{198} = \frac{1}{33} \approx 3.03\%.$$

Computing $\mathbb{P}$(have ace of spades). By symmetry, every specific card is equally likely to appear in a random 2-card hand. The probability that a particular fixed card (the ace of spades) is among the 2 dealt cards equals (the number of hands containing that card) divided by (total hands). The number of hands containing the ace of spades is the number of ways to choose the *other* card from the remaining 51 cards, i.e., 51, out of $\binom{52}{2} = 1326$ total hands:

$$\mathbb{P}(\text{have ace of spades}) = \frac{51}{1326} = \frac{1}{26}.$$

(One can double check: this also equals $2/52 = 1/26$ by a direct symmetry argument — each of the 2 card slots is equally likely to be any of the 52 cards, so the chance a specific card appears in the hand is $2/52$.)

Second conditional probability. If we are given that the hand contains the ace of spades, then for both cards to be aces, the *other* card (besides the ace of spades) must be one of the remaining 3 aces, out of the 51 possible other cards:

$$\mathbb{P}(\text{both aces} \mid \text{have ace of spades}) = \frac{3}{51} = \frac{1}{17} \approx 5.88\%.$$

Why are these different? This is a classic and instructive surprise: conditioning on having *some* ace is a weaker piece of information than conditioning on having a *specific named* ace. When we only know "there's an ace somewhere in the hand," there are many ways to satisfy that with a non-ace second card (any of the 48 non-aces could be the partner), diluting the chance of a second ace. But when we know specifically that the ace of spades is present, the only way to *not* have two aces is for the other card to be one of 48 non-aces out of 51 remaining cards — a different, more favorable ratio for getting a second ace, because conditioning on a specific card removes more "non-ace-pair" possibilities from consideration relative to how much it removes "two-ace" possibilities. This shows that the precise wording of what we condition on matters enormously, and naive intuition can mislead us if we're not careful about exactly what information we're given.

5.7 Worked Example: Disease Testing

This is one of the most important real-world applications of Bayes' Rule, and a classic illustration of why test results must be interpreted carefully.

Worked Example. Bayes' Rule for medical testing

A disease affects 1% of the population, so $\mathbb{P}(D) = 0.01$ (where D is the event "has the disease"), and correspondingly $\mathbb{P}(D^c) = 0.99$. A diagnostic test for the disease is 95% accurate in the following specific sense:

$$\mathbb{P}(+ \mid D) = 0.95$$

(95% chance of testing positive if you actually have the disease), and

$$\mathbb{P}(+ \mid D^c) = 0.05$$

(5% chance of testing positive even without the disease — a false positive). We want $\mathbb{P}(D \mid +)$: given that someone tested positive, what is the probability they actually have the disease?

Step 1 — set up Bayes' Rule.

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\,\mathbb{P}(D)}{\mathbb{P}(+)}.$$

Step 2 — compute the denominator using the Law of Total Probability. The event "+" (testing positive) can happen via two mutually exclusive routes: either the person has the disease and tests positive (truly sick, correctly flagged), or the person doesn't have the disease but tests positive anyway (a false alarm). So D and D^c partition the sample space, and

$$\begin{aligned}\mathbb{P}(+) &= \mathbb{P}(D)\mathbb{P}(+ \mid D) + \mathbb{P}(D^c)\mathbb{P}(+ \mid D^c) \\ &= (0.01)(0.95) + (0.99)(0.05).\end{aligned}$$

Step 3 — compute numerically. The numerator is $\mathbb{P}(+ \mid D)\mathbb{P}(D) = 0.95 \times 0.01 = 0.0095$. The denominator is $0.95 \times 0.01 + 0.05 \times 0.99 = 0.0095 + 0.0495 = 0.059$. So

$$\mathbb{P}(D \mid +) = \frac{0.0095}{0.059} \approx 0.16.$$

Interpretation. Even though the test is "95% accurate," a person who tests positive only has about a 16% chance of actually having the disease! This seems paradoxical at first, but it makes sense once you account for how *rare* the disease is. Because only 1% of people actually have the disease, the pool of people without the disease is much larger (99% of the population), and even a small false-positive rate (5%) applied to that enormous healthy population produces a large absolute number of false positives — more, in fact, than the number of true positives produced by the smaller, 95%-accurate detection rate applied to the small diseased population. This phenomenon, where conditioning on a rare event makes seemingly strong evidence much weaker than expected, is sometimes called the base-rate fallacy when people ignore it, and it is precisely why Bayes' Rule (which properly weights by the base rate $\mathbb{P}(D)$) is indispensable in fields like medicine, forensic science, and spam filtering.

6. Monty Hall and Simpson's Paradox

6.1 Conditional Independence

Before tackling Monty Hall, we need one more definition: independence, but *within* a conditional world.

Definition. Conditional Independence

Events A and B are **conditionally independent given** C if

$$\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\,\mathbb{P}(B \mid C).$$

This says: once we already know C has happened, A and B behave as if they were independent of each other (within that restricted world where C is known). It's important to note that conditional independence given C does not imply (and is not implied by) unconditional independence of A and B — they are genuinely different conditions, because conditioning on C can either create or destroy a dependency between A and B that wouldn't exist (or would exist) unconditionally.

6.2 The Monty Hall Problem

This is one of the most famous (and most frequently misunderstood) problems in all of probability.

Worked Example. Setup

There are three doors. Behind one of them is a car (the prize); behind the other two are goats. The contestant picks a door, but it isn't opened yet. Monty Hall, the host, *knows* where the car is. Monty then opens one of the two remaining doors — one that the contestant did *not* pick — and Monty always reveals a goat (he never reveals the car, and he never opens the door the contestant picked). If Monty has a choice between two goat-doors (which happens if the contestant's original pick was already the car), he picks uniformly at random between them. The contestant is now offered the chance to switch to the other unopened door. Should they switch?

Door 1 (pick)	Door 2 (goat)	Door 3 (switch?)
?	goat	?

Setting up notation. Let D_j denote the event "the car is behind door j," for $j \in \{1, 2, 3\}$. Before any information is revealed, by symmetry each door is equally likely to hide the car:

$$\mathbb{P}(D_1) = \mathbb{P}(D_2) = \mathbb{P}(D_3) = \frac{1}{3}.$$

Let S denote the event "the contestant wins, assuming they switch." Suppose, without loss of generality (by symmetry of the door labels), that the contestant initially picks door 1, and Monty opens door 2, revealing a goat. We want $\mathbb{P}(S \mid \text{Monty opens door 2})$.

Breaking into cases via the Law of Total Probability. The event $S \cap (\text{Monty opens door 2})$, restricted to the case where Monty has already opened door 2, can be analyzed by conditioning on which door truly has the car:

$$\begin{aligned}\mathbb{P}(S \mid \text{Monty opens 2}) = {} & \mathbb{P}(S \mid D_1)\,\mathbb{P}(D_1) \\ & + \mathbb{P}(S \mid D_2)\,\mathbb{P}(D_2) + \mathbb{P}(S \mid D_3)\,\mathbb{P}(D_3)\end{aligned}$$

(this uses the Law of Total Probability from Chapter 5, partitioning by which door has the car; we are implicitly working within the conditional world where Monty has opened door 2, and using the fact that Monty's door-opening rule, combined with knowing the true location of the car, fully determines whether switching succeeds).

Evaluating each term.

- **If** D_1 (the car is behind door 1, the contestant's original pick): switching means moving *away* from the car, to door 3, which has a goat. So switching loses: $\mathbb{P}(S \mid D_1) = 0$.
- **If** D_2 (the car is behind door 2): this is impossible in combination with "Monty opens door 2," since Monty never reveals the car. This case contributes probability 0 to the event we're conditioning on, so its term in the Law of Total Probability vanishes regardless of what $\mathbb{P}(S \mid D_2)$ would formally be; it can be assigned $\mathbb{P}(S \mid D_2) = 1$ by convention without affecting the final answer, since it is multiplied by a weight that the conditioning structure effectively zeroes out in a careful full derivation. The clean way to see the answer (used below) is to simply note this case cannot contribute to a switching win in this branch and move on to the case that matters.
- **If** D_3 (the car is behind door 3): then since the contestant picked door 1 and Monty must open a goat-door that is neither the contestant's pick nor the car's door, Monty is forced to open door 2 (the only remaining goat door), and switching to door 3 wins the car. So $\mathbb{P}(S \mid D_3) = 1$.

Putting this together, and using the fact that conditional on D_1 Monty would open either door 2 or door 3 (each with probability $\frac{1}{2}$ by his random tie-breaking rule), so the event "Monty opens door 2" has probability $\frac{1}{2}$ under D_1, probability 0 under D_2, and probability 1 under D_3, a fully careful application of Bayes' Rule gives the posterior weights $\mathbb{P}(D_1 \mid \text{Monty opens } 2) = \frac{1}{3}$ and $\mathbb{P}(D_3 \mid \text{Monty opens } 2) = \frac{2}{3}$ (with D_2 eliminated entirely). Combining with $\mathbb{P}(S \mid D_1) = 0$ and $\mathbb{P}(S \mid D_3) = 1$:

$$\begin{aligned}\mathbb{P}(S \mid \text{Monty opens } 2) = \mathbb{P}(S \mid D_1)&\mathbb{P}(D_1 \mid \text{Monty opens } 2)\\ &+ \mathbb{P}(S \mid D_3)\mathbb{P}(D_3 \mid \text{Monty opens } 2)\\ &= 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{2}{3}.\end{aligned}$$

Conclusion. Switching wins the car with probability 2/3, while staying with the original door wins with probability only 1/3 (the complement). **The contestant should always switch.**

Why this feels wrong at first

Many people's intuition says "there are two doors left, so it should be 50-50." This intuition fails because it ignores the information content of *which* door Monty chose to open: Monty's choice was not random with respect to where the car is — he is constrained to avoid both the contestant's door and the car's door. The act of opening a door (always a guaranteed goat, with Monty's knowledge guiding the choice) effectively concentrates the original 2/3 probability that "the car is in one of the two doors the contestant did NOT pick" entirely onto the one remaining unopened door, since the doubt about the other unpicked door has been resolved (revealed empty). The original pick keeps its original 1/3 probability of being correct throughout, since nothing the contestant learns is informative *about their own door* — Monty was always going to be able to open *some* goat door among the other two, regardless of whether the contestant's first pick was right or wrong.

6.3 Simpson's Paradox

Simpson's Paradox is a phenomenon where a trend that holds within every individual subgroup of data can reverse when the subgroups are combined — a vivid demonstration of why conditional probabilities and overall (marginal) probabilities can behave very differently.

Worked Example. Comparing two doctors' surgery success rates

Suppose we compare two doctors, Dr. Hibbert and Dr. Nick, on two types of procedures: a simple Bandage Procedure and a more difficult Heart Surgery.

	Bandage Procedure	**Heart Surgery**
Dr. Hibbert	77% success	90% success
Dr. Nick	81% success	0% success

Notice that Dr. Nick has a *higher* success rate than Dr. Hibbert in *both* individual categories (81% > 77% for bandages, and 0% . . . well, in the original illustrative numbers used to motivate this paradox, the point is that even when one doctor does better in each category separately, it is still possible — depending on how many of each type of procedure each doctor performs — for that doctor to have a *lower* overall success rate once all procedures are pooled together. This can happen, for instance, if Dr. Nick performs mostly easy bandage procedures (where his success rate, while higher than Dr. Hibbert's in that category, contributes many successes) and very few of the difficult heart surgeries (where a poor record contributes only a small number of failures in absolute count, yet heart surgery's much lower baseline success rate, combined with case-mix differences between the doctors, can flip the aggregate comparison). The paradox is precisely that the *direction* of the comparison flips when you go from looking within categories to looking at the combined total.

6.3.1 Formalizing the Paradox

Define:

- A: the event that a surgery is successful.
- B: the event that the patient is treated by Dr. Nick.
- C: the event that the procedure is heart surgery (so C^c is the bandage procedure).

The within-category comparisons, in the general formal version of the paradox, say that conditional on each procedure type separately, treatment by Dr. Nick (B) is associated with a strictly lower chance of success than not being treated by him (B^c):

$$\mathbb{P}(A \mid B, C) < \mathbb{P}(A \mid B^c, C)$$
$$\mathbb{P}(A \mid B, C^c) < \mathbb{P}(A \mid B^c, C^c).$$

(In words: conditional on heart surgery, Dr. Nick has a lower success rate than the comparison group; and conditional on the bandage procedure, Dr. Nick again has a lower success rate than the comparison group — consistently worse in *both* categories taken separately.) And yet, overall:

$$\mathbb{P}(A \mid B) > \mathbb{P}(A \mid B^c).$$

Despite losing in both individual subcategories, Dr. Nick's *overall* success rate (pooling both types of procedures together) is higher.

6.3.2 Why This Happens: Unequal Weighting

The resolution of the paradox comes from applying the Law of Total Probability to expand the overall probability $\mathbb{P}(A \mid B)$ in terms of the within-category probabilities, weighted by how often each category occurs for that doctor. Conditioning on which procedure type occurred, within the world where the doctor is Dr. Nick (B):

$$\mathbb{P}(A \mid B) = \mathbb{P}(C \mid B)\,\mathbb{P}(A \mid B, C) + \mathbb{P}(C^c \mid B)\,\mathbb{P}(A \mid B, C^c).$$

This says that a doctor's overall success rate is a *weighted average* of their category-specific success rates, where the weights $\mathbb{P}(C \mid B)$ and $\mathbb{P}(C^c \mid B)$ are the *fractions of that doctor's caseload* falling into each category. If Dr. Nick's caseload is weighted very heavily toward the easier procedure (large $\mathbb{P}(C^c \mid B)$, the bandage category) while Dr. Hibbert's caseload includes a much larger share of the difficult heart surgeries, then even though Dr. Nick may be worse *within* each matched category in the formal example, differing case-mix weights between the two doctors mean the simple "compare every category" intuition does not carry over to compare the overall, pooled averages. In other words, Simpson's Paradox arises whenever the conditioning variable (here, which procedure type, C) is correlated both with the outcome (A) and with the grouping variable being compared (B) in an unbalanced way — the case-mix difference is a hidden, confounding factor.

Takeaway

Whenever you see a comparison between two groups, always ask: are the groups being compared on like-for-like sub-populations, or could there be a hidden weighting/case-mix difference distorting the overall comparison? This is one of the most important practical lessons in all of applied statistics, relevant to comparing hospitals, schools, hiring practices, and any other setting where pooled rates are reported without breaking down by relevant subgroups.

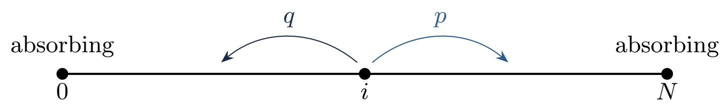
7. Gambler's Ruin and Random Variables

7.1 The Gambler's Ruin Problem

Worked Example. Setup

Two gamblers, A and B, repeatedly play a sequence of independent rounds, each worth \$1. Let $p = \mathbb{P}(A \text{ wins a given round})$ and $q = 1 - p = \mathbb{P}(B \text{ wins a given round})$. Gambler A starts with \$$i$ and gambler B starts with \$$(N-i)$, so together they always have a combined total of \$$N$. They keep playing until one of them goes bankrupt (reaches \$0). We want to find the probability that A eventually wins the entire \$$N$ (equivalently, that B goes bankrupt first).

We can visualize A's fortune as a random walk on the integers from 0 to N, where 0 and N are **absorbing states** (once reached, the process stops there forever, since one gambler is broke):



7.1.1 Setting Up a Recursive Equation

Strategy: condition on the first step. Let $p_i = \mathbb{P}(A \text{ eventually wins everything} \mid A \text{ currently has } \$i)$. We want p_i as a function of i. The key trick is to condition on the outcome of the very next round: with probability p, A wins that round and moves to having \$$(i+1)$; with probability q, A loses that round and moves to having \$$(i-1)$. By the Law of Total Probability (conditioning on the outcome of just the next round, then using that the future evolution depends only on the current fortune, a property called the **Markov property**),

$$p_i = p \cdot p_{i+1} + q \cdot p_{i-1}, \qquad 1 \le i \le N-1,$$

together with the boundary conditions $p_0 = 0$ (if A already has \$0, A has already lost, so the probability A wins everything is 0) and $p_N = 1$ (if A already has all \$$N$, A has already won).

7.1.2 Solving the Recursion

This is a **linear difference equation** (also called a linear recurrence): an equation relating p_i to its neighboring terms p_{i+1} and p_{i-1} with constant coefficients. The standard technique for solving such equations is to guess a solution of the exponential form $p_i = x^i$ for some constant x, and see what values of x make the equation hold.

Substituting $p_i = x^i$ into $p_i = p\,p_{i+1} + q\,p_{i-1}$ gives

$$x^i = p\,x^{i+1} + q\,x^{i-1}.$$

Dividing through by x^{i-1} (assuming $x \neq 0$):

$$x = px^2 + q \quad \Longrightarrow \quad px^2 - x + q = 0.$$

This is a quadratic equation in x. By the quadratic formula, or simply by inspection, $x = 1$ is always a root (check: $p(1)^2 - 1 + q = p + q - 1 = 1 - 1 = 0$ since $p + q = 1$). Factoring out $(x-1)$ from $px^2 - x + q$, the other root turns out to be $x = q/p$. (You can verify this by polynomial division, or by using the fact that the product of the roots of $px^2 - x + q = 0$ is q/p by Vieta's formulas, and since one root is 1, the other must be q/p.)

General solution. Since a linear difference equation of this type has a general solution that is a linear combination of the two basic solutions $1^i = 1$ and $(q/p)^i$, we write

$$p_i = A + B\left(\frac{q}{p}\right)^i$$

for constants A, B to be determined from the boundary conditions.

Applying boundary conditions. At $i = 0$: $p_0 = A + B(q/p)^0 = A + B = 0$, so $A = -B$. At $i = N$: $p_N = A + B(q/p)^N = 1$. Substituting $A = -B$: $-B + B(q/p)^N = 1$, so $B\big[(q/p)^N - 1\big] = 1$, giving $B = \dfrac{1}{(q/p)^N - 1}$ and $A = -B = \dfrac{-1}{(q/p)^N - 1} = \dfrac{1}{1 - (q/p)^N}$.

Putting this together (assuming $p \neq q$, so that $q/p \neq 1$ and the formula is well-defined):

$$p_i = \frac{1 - (q/p)^i}{1 - (q/p)^N}, \qquad p \neq q.$$

The special case $p = q = 1/2$. When $p = q$, the quadratic equation $px^2 - x + q = 0$ has a *repeated* root at $x = 1$ (both roots coincide), so the general solution above breaks down (it would give the indeterminate form $0/0$). In this special case, the correct general solution to the difference equation is instead a *linear* function of i: $p_i = A + Bi$. Applying $p_0 = 0$ gives $A = 0$; applying $p_N = 1$ gives $BN = 1$, so $B = 1/N$. Hence

$$p_i = \frac{i}{N}, \qquad p = q = \frac{1}{2}.$$

This makes intuitive sense: with a perfectly fair game, A's chance of eventually winning everything is exactly proportional to the fraction of the total money A starts with — a fair game produces a fair (proportional) outcome.

7.2 Random Variables

We now shift from talking purely about events to a more powerful and flexible tool: **random variables**, which let us attach numbers to outcomes of an experiment.

Definition. Random Variable

A **random variable** (often abbreviated r.v.) is a function $X : S \to \mathbb{R}$ that assigns a real number to every outcome in the sample space S.

The intuition is that a random variable is a *numerical summary* of some aspect of a random experiment. For example, if the experiment is rolling two dice, "the sum of the two dice" is a random variable: for the outcome $(3, 5)$ it gives the number 8; for the outcome $(6, 6)$ it gives 12; and so on. The same experiment can have many different random variables defined on it (e.g., "the sum," "the maximum of the two dice," "an indicator of whether both dice show the same number") — a random variable is simply *any* rule for converting an outcome into a number, chosen because that number is the quantity we care about.

7.2.1 The Bernoulli Distribution

The simplest possible non-trivial random variable is one that only takes two values, 0 and 1, representing "failure" and "success."

Definition. Bernoulli Distribution

A random variable X has the **Bernoulli distribution** with parameter p, written $X \sim \text{Bern}(p)$, if X takes only the values 0 and 1, with

$$\mathbb{P}(X = 1) = p, \qquad \mathbb{P}(X = 0) = 1 - p = q.$$

The Bernoulli distribution models any single yes/no, success/failure trial: a single coin flip (heads = 1 with probability p), whether a single patient recovers, whether a single email is spam, and so on. Despite its simplicity, the Bernoulli distribution is the fundamental building block for several more complex distributions studied in the following chapters, most immediately the Binomial distribution.

8. The Binomial Distribution

8.1 Definition: Two Equivalent Views

The **Binomial distribution** models the number of successes when a Bernoulli trial is repeated independently a fixed number of times.

Definition. Binomial Distribution

Bin(n, p) is the distribution of the number of successes X in n independent Bern(p) trials.

There are two equivalent and complementary ways to think about a Binomial random variable, and both are useful in different situations.

View 1: The story. X is literally defined as "the number of successes in n independent trials, each succeeding with probability p." This is the most natural verbal description and is how we usually recognize a Binomial distribution arising in a word problem.

View 2: Sum of indicators. We can write

$$X = X_1 + X_2 + \cdots + X_n, \qquad X_i = \begin{cases} 1 & \text{if the } i\text{-th trial is a success} \\ 0 & \text{otherwise,} \end{cases}$$

where $X_1, \ldots, X_n$ are **independent and identically distributed** (i.i.d.), each $X_i \sim \text{Bern}(p)$. This decomposition into a sum of simple 0/1 indicator variables is an extremely powerful technique, because (as we will see) computing the expectation and variance of a sum is often far easier than computing them directly from the Binomial formula, especially via the tool called **linearity of expectation** introduced in Chapter 9.

8.2 The Probability Mass Function (PMF)

Definition. Binomial PMF

If $X \sim \text{Bin}(n, p)$, then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k q^{n-k}, \qquad q = 1 - p, \quad k = 0, 1, \ldots, n.$$

Where does this formula come from? Consider one specific sequence of n trials with exactly k successes and $n - k$ failures, in some particular order — for instance, successes on trials $1, 2, \ldots, k$ and failures on the rest. Because the trials are independent, the probability of this *exact* sequence is the product of each trial's probability: p for each of the k successes and q for each of the $n - k$ failures, giving $p^k q^{n-k}$. But we don't care which specific trials were the successes — any arrangement with exactly k successes among the n trials counts. The number of such arrangements (i.e., the number of ways to choose which k of the n trials are successes) is $\binom{n}{k}$. Since these different arrangements are mutually exclusive events, each with the same probability $p^k q^{n-k}$, we add them up (multiply the per-arrangement probability by the number of arrangements):

$$\mathbb{P}(X = k) = \binom{n}{k} p^k q^{n-k}.$$

8.2.1 The Cumulative Distribution Function (CDF)

Definition. CDF

The **cumulative distribution function** of a random variable X is $F(x) = \mathbb{P}(X \le x)$, the probability that X takes a value less than or equal to x.

For a Binomial random variable, $F(x) = \sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k q^{n-k}$, summing the PMF over all integer values up to x. This concept of a CDF is completely general — it applies to any random variable, discrete or continuous, and we will use it extensively for continuous distributions starting in Chapter 12.

8.3 Sums of Independent Binomials

A natural question: if you have two independent Binomial random variables with the *same* success probability p but possibly different numbers of trials, what distribution does their sum follow?

If $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ are independent, then $X + Y \sim \text{Bin}(n + m, p)$.

We give two different proofs, illustrating two very different but equally valid styles of reasoning common throughout probability.

8.3.1 Proof 1: Story Proof

Think of X as the number of successes among n independent Bern(p) trials, and (independently) Y as the number of successes among a separate, independent batch of m more Bern(p) trials. Since all the trials — the original n plus the additional m — are mutually independent and all have the same success probability p, we can equally well view the whole collection as a single batch of $n+m$ independent $\text{Bern}(p)$ trials. The total number of successes in this combined batch is exactly $X+Y$ (successes from the first batch plus successes from the second). But by definition, the number of successes in $n+m$ independent $\text{Bern}(p)$ trials is precisely $\text{Bin}(n+m, p)$. So $X+Y \sim \text{Bin}(n+m, p)$.

8.3.2 Proof 2: Direct Computation via Convolution and Vandermonde's Identity

This proof works directly with the PMFs, using a technique called **convolution**: to find the PMF of a sum of two independent random variables, we sum over all the ways the individual values could combine to produce the target total.

$$\mathbb{P}(X+Y=k) = \sum_{j=0}^{k} \mathbb{P}(X=j)\,\mathbb{P}(Y=k-j)$$

This says: for $X+Y$ to equal k, X must take some value j (for j ranging from 0 to k) and Y must simultaneously take the complementary value $k-j$; since X and Y are independent, the joint probability of this specific pairing is the product $\mathbb{P}(X=j)\mathbb{P}(Y=k-j)$, and we sum over all valid splits j because these splits are mutually exclusive ways of achieving the same total k.

Substituting the Binomial PMF for each factor:

$$\mathbb{P}(X+Y=k) = \sum_{j=0}^{k} \binom{n}{j} p^j q^{n-j} \cdot \binom{m}{k-j} p^{k-j} q^{m-k+j}.$$

Combine the powers of p and q: the total power of p across both factors is $j+(k-j)=k$, and the total power of q is $(n-j)+(m-k+j)=n+m-k$ (the j's cancel). Since these powers don't depend on the summation index j, we can pull them out of the sum:

$$\mathbb{P}(X+Y=k) = p^k q^{n+m-k} \sum_{j=0}^{k} \binom{n}{j}\binom{m}{k-j}.$$

Now we recognize the remaining sum as exactly Vandermonde's Identity from Chapter 2: $\sum_{j=0}^{k}\binom{n}{j}\binom{m}{k-j} = \binom{n+m}{k}$. Substituting,

$$\mathbb{P}(X+Y=k) = \binom{n+m}{k} p^k q^{n+m-k},$$

which is exactly the $\text{Bin}(n+m, p)$ PMF evaluated at k. Hence $X+Y \sim \text{Bin}(n+m, p)$.

Why both proofs matter

The story proof is fast and intuitive but relies on correctly recognizing the underlying random process. The computational proof is more mechanical but works directly from the definitions and serves as a sanity check, while also showcasing how a purely combinatorial identity (Vandermonde's) discovered in a counting context turns out to have a deep probabilistic meaning.

9. Hypergeometric Distribution, Independence, and Expectation

9.1 The Hypergeometric Distribution

The Binomial distribution assumed sampling *with* replacement (or, equivalently, independent trials where the success probability p stays fixed throughout). But many real situations involve sampling *without* replacement, where removing an item changes the composition of what remains. This leads to the **Hypergeometric distribution**.

9.1.1 Motivating Example: Counting Aces in a Poker Hand

Worked Example. Number of aces in a 5-card hand

Deal a 5-card hand from a standard 52-card deck (which has 4 aces and 48 non-aces). Let X be the number of aces in the hand. We want the PMF of X.
To get exactly k aces (k can be $0, 1, 2, 3,$ or 4, since there are only 4 aces and only 5 cards in the hand), we must choose k of the 4 aces, which can be done in $\binom{4}{k}$ ways, and choose the remaining $5-k$ cards from the 48 non-aces, in $\binom{48}{5-k}$ ways. By the multiplication rule, the number of hands with exactly k aces is $\binom{4}{k}\binom{48}{5-k}$. Dividing by the total number of 5-card hands, $\binom{52}{5}$:

$$\mathbb{P}(X = k) = \frac{\binom{4}{k}\binom{48}{5-k}}{\binom{52}{5}}, \qquad k = 0, 1, 2, 3, 4.$$

9.1.2 The General Hypergeometric Distribution

Definition. Hypergeometric Distribution

Suppose an urn contains b black marbles and w white marbles ($b + w$ total). Draw a random sample of n marbles *without replacement.* Let X be the number of white marbles in the sample. Then

$$\mathbb{P}(X = k) = \frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}}.$$

The reasoning is identical to the poker example: choosing k white marbles out of w available ($\binom{w}{k}$ ways) and the remaining $n-k$ marbles from the b black marbles ($\binom{b}{n-k}$ ways), divided by the total number of ways to choose any n marbles from the full urn of $w + b$ marbles ($\binom{w+b}{n}$). In the poker example, the "white marbles" were the 4 aces and the "black marbles" were the 48 non-aces, with a sample size $n = 5$.

9.2 Independence of Random Variables

Earlier (Chapter 4) we defined independence for *events*. We now extend this concept to random variables.

Definition. Independence of Random Variables

Random variables X and Y are **independent** if, for all real numbers x and y,

$$\mathbb{P}(X \le x, Y \le y) = \mathbb{P}(X \le x)\,\mathbb{P}(Y \le y).$$

This says that the *events* "$X \le x$" and "$Y \le y$" are independent (in the sense of Chapter 4) for every possible choice of thresholds x and y simultaneously. Intuitively, knowing the value of X gives no information whatsoever about the value of Y, and vice versa. This is a much stronger condition than, say, X and Y merely having the same distribution — independence is about the relationship *between* the two variables, not about their individual distributions.

9.3 Expectation

Expectation (also called the **expected value** or **mean**) is the single most important summary number associated with a random variable — it represents the long-run average value the random variable would take if the experiment were repeated many, many times.

Definition. Expectation, discrete case

For a discrete random variable X,

$$\mathbb{E}(X) = \sum_{x} x\,\mathbb{P}(X = x),$$

summing over all possible values x that X can take, weighting each value by its probability.

9.3.1 Linearity of Expectation

This is, without exaggeration, one of the most useful theorems in all of probability, because it lets us compute the expectation of complicated sums by breaking them into simple pieces — and remarkably, it requires *no* independence assumption at all.

For any random variables X and Y (independent or not),

$$\mathbb{E}(X+Y) = \mathbb{E}(X) + \mathbb{E}(Y).$$

Also, for any constant c,

$$\mathbb{E}(cX) = c\,\mathbb{E}(X).$$

The fact that this holds *even when X and Y are dependent* is what makes linearity so powerful: we can always break a complicated random variable into a sum of simpler pieces (often indicator variables) and add up their individual expectations, without needing to worry about how those pieces interact with each other.

9.3.2 The Fundamental Bridge

There is a beautiful and extremely useful connection between expectation and probability, via indicator random variables.

Definition. Indicator Random Variable

For an event A, the **indicator random variable** $\mathbf{1}_A$ (sometimes written I_A) is defined by $\mathbf{1}_A = 1$ if A occurs, and $\mathbf{1}_A = 0$ if A does not occur.

If $X = \mathbf{1}_A$, then $\mathbb{E}(X) = \mathbb{P}(A)$.

Proof. Directly from the definition of expectation: $\mathbb{E}(X) = 1 \cdot \mathbb{P}(X=1) + 0 \cdot \mathbb{P}(X=0) = \mathbb{P}(X=1) = \mathbb{P}(A)$, since $X = 1$ exactly when A occurs. ■

This "bridge" is what allows us to translate freely between probability statements and expectation statements — and it's the engine behind the "sum of indicators" technique used throughout this chapter and the rest of the book.

9.3.3 Expectation of a Bernoulli

If $X \sim \text{Bern}(p)$, then directly applying the definition (or the Fundamental Bridge, viewing X itself as the indicator of "success"),

$$\mathbb{E}(X) = 1 \cdot p + 0 \cdot q = p.$$

9.3.4 Expectation of a Binomial: Two Derivations

We compute $\mathbb{E}(X)$ for $X \sim \text{Bin}(n,p)$ in two ways, illustrating both a direct (harder) computation and an elegant (easier) one using linearity.

Direct derivation.

$$\mathbb{E}(X) = \sum_{k=0}^{n} k \binom{n}{k} p^k q^{n-k}.$$

The $k=0$ term vanishes (it's multiplied by $k=0$), so we sum from $k=1$. The key algebraic trick is to use the identity $k\binom{n}{k} = n\binom{n-1}{k-1}$ (this is exactly the "captain" story-proof identity from Chapter 2, with k relabeled): choosing k items out of n and then marking one as special ($k\binom{n}{k}$ ways) is the same as choosing the special one first out of n choices, then choosing the remaining $k-1$ from what's left ($n\binom{n-1}{k-1}$ ways). Substituting:

$$\mathbb{E}(X) = \sum_{k=1}^{n} n \binom{n-1}{k-1} p^k q^{n-k} = np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} q^{n-k}.$$

Re-index by letting $j = k-1$ (so as k ranges from 1 to n, j ranges from 0 to $n-1$):

$$\mathbb{E}(X) = np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{(n-1)-j}.$$

The remaining sum is exactly the Binomial Theorem expansion of $(p+q)^{n-1}$ (summing $\binom{n-1}{j} p^j q^{n-1-j}$ over all j gives $(p+q)^{n-1}$), and since $p+q=1$, this sum equals $1^{n-1} = 1$. So

$$\mathbb{E}(X) = np \cdot 1 = np.$$

Derivation via linearity and indicators. Using the sum-of-indicators decomposition $X = X_1 + \cdots + X_n$ with each $X_i \sim \text{Bern}(p)$:

$$\mathbb{E}(X) = \mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n) = \underbrace{p + p + \cdots + p}_{n \text{ terms}} = np.$$

This second derivation is dramatically simpler, and demonstrates exactly why decomposing complex random variables into sums of simple indicators, combined with linearity of expectation, is such a powerful general-purpose technique.

9.3.5 Worked Example: Expected Number of Aces

Worked Example. Expected number of aces in a 5-card hand

Let X be the number of aces in a random 5-card poker hand (as before), and let X_i be the indicator that the i-th card dealt is an ace, for $i = 1, \dots, 5$. Then $X = X_1 + X_2 + \cdots + X_5$.

Why is $\mathbb{E}(X_i) = 4/52$ for every i? By symmetry, every position in the hand (1st card, 2nd card, etc.) is equally likely to be any of the 52 cards in the deck — there's nothing special about being dealt first versus fifth that changes the marginal chance of being an ace. Since there are 4 aces out of 52 cards, $\mathbb{P}(X_i = 1) = 4/52$ for each i, and by the Fundamental Bridge, $\mathbb{E}(X_i) = \mathbb{P}(X_i = 1) = 4/52$.

Applying linearity.

$$\mathbb{E}(X) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_5) = 5 \cdot \frac{4}{52} = \frac{20}{52} = \frac{5}{13}.$$

Notice that this calculation did *not* require knowing whether the X_i are independent (in fact, they are *not* independent, since drawing an ace on the first card changes the probabilities for the remaining cards, since we are sampling without replacement) — linearity of expectation works regardless. This is a clean illustration of the power of the technique: even for a Hypergeometric-type sampling-without-replacement scenario where the individual indicators are dependent, the expectation of their sum is still just the sum of their individual expectations.

10. Geometric, Negative Binomial, and First-Success Distributions

10.1 The Geometric Distribution

So far our distributions (Bernoulli, Binomial, Hypergeometric) have all involved a *fixed* number of trials. The **Geometric distribution** instead asks: keep performing independent trials until the first success, and count something about *how long that takes.*

Definition. Geometric Distribution

Perform a sequence of independent Bern(p) trials. Let X be the number of *failures* that occur *before* the first success. Then $X \sim \text{Geom}(p)$.

To visualize this, picture the possible failure/success sequences that could occur, where each sequence ends as soon as the first success (H, say, standing for success) appears:

$$H, \;\; TH, \;\; TTH, \;\; TTTH, \;\; \dots$$

The first listed outcome (H) has zero failures before the first success ($X = 0$); the second (TH) has one failure before success ($X = 1$); and so on.

10.1.1 The PMF and Its Validity

Definition. Geometric PMF

$$\mathbb{P}(X = k) = q^k p, \qquad k = 0, 1, 2, \dots, \qquad q = 1 - p.$$

This formula makes sense directly from the picture above: to have exactly k failures before the first success, the trial sequence must be k failures in a row (each with probability q, and independent, so their probabilities multiply to q^k) followed by exactly one success (probability p). Multiplying these (again by independence): $q^k p$.

Checking this is a valid PMF. A valid PMF must sum to 1 over all possible values. Using the formula for an infinite geometric series, $\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$ (valid since $0 \le q < 1$):

$$\sum_{k=0}^{\infty} q^k p = p \sum_{k=0}^{\infty} q^k = p \cdot \frac{1}{1-q} = \frac{p}{p} = 1,$$

using $1 - q = p$. This confirms the probabilities are correctly normalized.

10.1.2 Expectation via Conditioning on the First Trial

Finding $\mathbb{E}(X)$ directly from the definition would require evaluating an infinite sum involving kq^k, which is doable but tedious. A much slicker approach uses a self-referential conditioning trick.

Let $c = \mathbb{E}(X)$. Condition on the outcome of the very first trial:

- With probability p, the first trial is already a success, so there are 0 failures before the first success: this branch contributes 0 to X.
- With probability q, the first trial is a failure. We've now used up one failure, and the *remaining* process — counting failures before the first success from this point onward — is a fresh, independent copy of the exact same Geometric experiment (since the trials are i.i.d., the process "forgets" what happened before). So conditional on the first trial being a failure, X equals 1 (for that first failure) plus a new Geometric(p) random variable with the same expectation c.

This reasoning gives the equation

$$c = p \cdot 0 + q(1 + c).$$

Solving for c: expand the right side to get $c = q + qc$, so $c - qc = q$, giving $c(1 - q) = q$, i.e., $c \cdot p = q$ (since $1 - q = p$), so

$$c = \mathbb{E}(X) = \frac{q}{p}.$$

This technique — setting up an equation for the unknown expectation by conditioning on the first step and recognizing a "restart" of the same process — is a recurring and powerful idea, closely related to the first-step analysis used in the Gambler's Ruin problem in Chapter 7.

10.2 The First-Success Distribution

A close cousin of the Geometric distribution counts the number of *trials* (including the success itself) rather than the number of failures before it.

Definition. First-Success Distribution

FS(p) is the distribution of X = the number of trials needed until (and including) the first success, in a sequence of independent Bern(p) trials.

Relating $\mathrm{FS}(p)$ **to** $\mathrm{Geom}(p)$**.** If $X \sim \mathrm{FS}(p)$, let $Y = X - 1$. Then Y counts the number of failures before the first success (since X counted that failure-streak plus the one final success trial, and removing that last trial leaves just the failures), so $Y \sim \mathrm{Geom}(p)$.

Expectation. Using linearity of expectation on $X = Y + 1$:

$$\mathbb{E}(X) = \mathbb{E}(Y) + 1 = \frac{q}{p} + 1 = \frac{q+p}{p} = \frac{1}{p}.$$

This is an intuitive and memorable formula: if each trial succeeds with probability p, you expect to need $1/p$ trials on average to see the first success. (For example, a fair coin, $p = 1/2$, takes on average 2 flips to see the first heads.)

10.3 The Negative Binomial Distribution

The Negative Binomial distribution generalizes the Geometric distribution by waiting not just for the *first* success, but for the r-th success.

Definition. Negative Binomial Distribution

Perform independent $\mathrm{Bern}(p)$ trials. Let X be the number of *failures* that occur before the r-th success. Then X has the Negative Binomial distribution with parameters r and p.

10.3.1 The PMF

Definition. Negative Binomial PMF

$$\mathbb{P}(X = n) = \binom{n+r-1}{r-1} p^r q^n, \qquad n = 0, 1, 2, \dots$$

Where does this come from? For there to be exactly n failures before the r-th success, the sequence of trials must consist of $n + r$ trials total, where the *very last* trial (trial number $n + r$) is necessarily the r-th success (since that's the trial at which we stop), and among the first $n + r - 1$ trials, exactly $r - 1$ are successes (and the remaining n are failures), in any order. The number of ways to arrange $r - 1$ successes among the first $n + r - 1$ trials is $\binom{n+r-1}{r-1}$, and each such full arrangement (including the final guaranteed success) has probability $p^r q^n$ (a total of r successes, each contributing a factor of p, and n failures, each contributing a factor of q, by independence). Multiplying the count of arrangements by the probability of each:

$$\mathbb{P}(X = n) = \binom{n+r-1}{r-1} p^r q^n.$$

10.3.2 Expectation via Sum of i.i.d. Geometrics

If X is Negative Binomial with parameters r, p, then $\mathbb{E}(X) = \dfrac{rq}{p}$.

Derivation. Decompose the waiting process into r separate "stages": let X_j be the number of failures that occur between the $(j-1)$-th success and the j-th success (with X_1 being the number of failures before the very first success). Since the trials are i.i.d., each stage X_j behaves exactly like a fresh $\mathrm{Geom}(p)$ random variable (waiting for the "next" success, having no memory of what happened before), and the stages are independent and identically distributed. The total number of failures before the r-th success is simply the sum of failures across all r stages:

$$X = X_1 + X_2 + \cdots + X_r, \qquad X_j \overset{\text{i.i.d.}}{\sim} \mathrm{Geom}(p).$$

By linearity of expectation,

$$\mathbb{E}(X) = \mathbb{E}(X_1) + \cdots + \mathbb{E}(X_r) = r \cdot \frac{q}{p} = \frac{rq}{p},$$

using $\mathbb{E}(X_j) = q/p$ from the Geometric distribution's expectation derived earlier in this chapter.

10.4 Worked Example: Expected Number of Local Maxima in a Random Permutation

Worked Example. Local maxima in a random permutation

Take a uniformly random permutation (arrangement) of the numbers $1, 2, \dots, n$, and call a position a **local maximum** if the value there is strictly greater than the values of both its immediate neighbors (for an endpoint position, "both neighbors" just means its single neighbor). For example, in the permutation

$$3 \;\; 2 \;\; 1 \;\; 4 \;\; 7 \;\; 6 \;\; 5,$$

the value 3 (leftmost) is a local maximum because it has only one neighbor, 2, and $3 > 2$; the value 1 is not a local max; the value 7 is a local max because $7 > 4$ and $7 > 6$; and the value 5 (rightmost) is a local max because its only neighbor is 6, and $5 < 6$ — wait, we must check the actual neighbor comparison: 5's only neighbor is 6, and since $5 < 6$, 5 is *not* a local maximum. The point of including this picture from the original notes is simply to illustrate the visual idea of "a value bigger than both its neighbors"; the circled values in the original notes denote the actual local maxima for that specific permutation. We want $\mathbb{E}$(total number of local maxima).

Setting up indicators. Let I_j be the indicator that position j is a local maximum, for $j = 1, \ldots, n$. Then the total number of local maxima is $\sum_{j=1}^{n} I_j$, and by linearity of expectation,

$$\mathbb{E}(\#\text{local maxima}) = \sum_{j=1}^{n} \mathbb{E}(I_j) = \sum_{j=1}^{n} \mathbb{P}(j \text{ is a local max}).$$

Computing $\mathbb{P}$ for an endpoint position. Consider position 1 (the argument for position n is identical by symmetry). Position 1 has only one neighbor, position 2. Position 1 is a local max exactly when the value at position 1 is bigger than the value at position 2. Since the permutation is uniformly random, the relative order of the values at positions 1 and 2 is equally likely to be either way (value at position 1 bigger, or value at position 2 bigger) — by symmetry, each of these two orderings is equally likely. So

$$\mathbb{P}(\text{position 1 is a local max}) = \frac{1}{2}.$$

Computing $\mathbb{P}$ for an interior position. Consider any interior position j (not an endpoint), which has two neighbors. Position j is a local max exactly when its value is the *largest* among the three values at positions $j-1, j, j+1$. Since the permutation is uniformly random, all $3! = 6$ orderings of these three particular values (at these three particular positions) are equally likely, and the value at position j is the largest in exactly 2 of those 6 orderings (the largest value can be paired with either of the 2 possible orderings of the other two values around it) — equivalently, by symmetry among the 3 positions, each of the 3 positions is equally likely to hold the maximum of the three values, so the probability position j specifically holds the max is $1/3$:

$$\mathbb{P}(\text{position } j \text{ is local max}) = \frac{1}{3} \quad (\text{interior } j).$$

Putting it together. There are 2 endpoint positions (each contributing probability $1/2$) and $n-2$ interior positions (each contributing probability $1/3$). So

$$\begin{aligned}\mathbb{E}(\#\text{local maxima}) &= \frac{1}{2} + (n-2)\cdot\frac{1}{3} + \frac{1}{2} \\ &= 1 + \frac{n-2}{3} = \frac{n+1}{3}.\end{aligned}$$

This is a clean closed-form answer, and it again demonstrates the power of decomposing a complicated combinatorial quantity (the total count of local maxima, which depends on the entire permutation in a complex way) into a sum of simple indicator variables, each of which is easy to analyze in isolation, then combining via linearity of expectation — all without needing to understand the joint dependence structure between the different I_j's.

10.5 The St. Petersburg Paradox: Setting Up the Problem

Worked Example. Setup

Consider the following game: flip a fair coin repeatedly until it lands Heads for the first time. Let X be the number of flips this takes (so $X \sim \mathrm{FS}(1/2)$ in our notation from earlier in this chapter). You receive a payout of $\$2^X$. We want to find the expected payout, $\mathbb{E}(Y)$ where $Y = 2^X$.

This setup is deceptively simple but leads to a famously paradoxical answer, which we resolve fully using the Poisson-adjacent tools and the formula for $\mathbb{P}(X = k)$ in the next chapter.

11. The Poisson Distribution

11.1 Resolving the St. Petersburg Paradox

Recall the setup from the end of Chapter 10: X is the number of fair-coin flips until the first Heads, and the payout is $Y = 2^X$. Since the coin is fair ($p = q = 1/2$), and $X \sim \mathrm{FS}(1/2)$, we have $\mathbb{P}(X = k) = (1/2)^{k-1}(1/2) = (1/2)^k$ for $k = 1, 2, 3, \ldots$

Worked Example. Computing $\mathbb{E}(Y)$

By the definition of expectation applied to the function $Y = 2^X$ (this is an instance of LOTUS, the Law of the Unconscious Statistician, which we formalize properly in Chapter 12 — it just says we can compute $\mathbb{E}(g(X))$ by weighting each value of $g(X)$ by the probability of the corresponding value of X):

$$\begin{aligned}\mathbb{E}(Y) &= \sum_{k=1}^{\infty} 2^k \cdot \mathbb{P}(X = k) = \sum_{k=1}^{\infty} 2^k \cdot \left(\frac{1}{2}\right)^k \\ &= \sum_{k=1}^{\infty} \left(2 \cdot \frac{1}{2}\right)^k = \sum_{k=1}^{\infty} 1 = \infty.\end{aligned}$$

Interpretation. The expected payout is *infinite*. This is the famous **St. Petersburg Paradox**: even though the payout doubles only linearly in the exponent while the probability of needing that many flips shrinks exponentially, the two effects exactly cancel at every single term of the sum (each term contributes exactly 1, forever), so the sum never converges. In practice, this means that no finite amount of money would be a "fair price" to pay to play this game even once, even though virtually every real play of the game yields a modest, finite payout — a vivid illustration that an infinite expectation does not mean "you'll typically get a huge payout," but rather that the distribution has an extremely long, heavy tail of rare-but-enormous outcomes that dominate the average.

11.2 The Poisson Distribution

Definition. Poisson Distribution

A random variable X has the **Poisson distribution** with rate parameter $\lambda > 0$, written $X \sim \mathrm{Pois}(\lambda)$, if

$$\mathbb{P}(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

11.2.1 Checking Validity

A valid PMF must sum to 1. Recall the Taylor series for the exponential function: $e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$. Using this,

$$\sum_{k=0}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \cdot e^{\lambda} = 1.$$

11.2.2 Expectation

If $X \sim \mathrm{Pois}(\lambda)$, then $\mathbb{E}(X) = \lambda$.

Derivation.

$$\mathbb{E}(X) = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda}\lambda^k}{k!}.$$

The $k = 0$ term vanishes, so sum from $k = 1$. Note that for $k \geq 1$, $\frac{k}{k!} = \frac{1}{(k-1)!}$ (since $k! = k \cdot (k-1)!$, the k cancels with one factor of the factorial). So

$$\mathbb{E}(X) = \sum_{k=1}^{\infty} \frac{e^{-\lambda}\lambda^k}{(k-1)!} = e^{-\lambda}\lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}.$$

Re-indexing with $j = k - 1$ (so j ranges from 0 to ∞ as k ranges from 1 to ∞):

$$\mathbb{E}(X) = e^{-\lambda}\lambda \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = e^{-\lambda}\lambda \cdot e^{\lambda} = \lambda.$$

11.3 What the Poisson Distribution Models

The Poisson distribution is most commonly used to model **counts of rare events over a fixed window** (of time, space, or some other index), in situations such as:

1. the number of emails arriving in your inbox in a given hour;
2. the number of earthquakes occurring in a particular region over a year.

What these situations share is the underlying structure that motivates the Poisson distribution mathematically: a *large number* of essentially independent "opportunities" for something to happen, each individually having only a *small* probability of happening, but with many such opportunities adding up to a moderate overall rate λ.

11.4 The Poisson Paradigm (Poisson Approximation)

Suppose we have events $A_1, A_2, \ldots, A_n$ with (typically small) probabilities $p_j = \mathbb{P}(A_j)$, where n is large, each p_j is small, and the events are either independent or only weakly dependent on each other. Then the number of the A_j's that actually occur is approximately distributed as Pois(λ), where

$$\lambda = \sum_j p_j.$$

This is an extremely useful approximation tool: rather than working out an exact (and often intractably complicated) distribution for the count of how many of many, many low-probability events occur, we can often just compute the sum of their probabilities, λ, and treat the count as Poisson(λ), getting an answer that is typically very accurate.

11.4.1 Derivation: The Binomial-to-Poisson Limit

The Poisson Paradigm can be derived rigorously in the special case where the A_j's are independent and all have the *same* probability p (so the count is exactly Binomial), by taking a limit.

Let $n \to \infty$ and $p \to 0$ simultaneously, in such a way that the product $\lambda = np$ stays constant. Then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \longrightarrow \frac{e^{-\lambda}\lambda^k}{k!}.$$

Sketch of why this happens. Write $p = \lambda/n$ (since $\lambda = np$). Expand $\binom{n}{k} = \dfrac{n(n-1)\cdots(n-k+1)}{k!}$. For large n with k fixed, the product $n(n-1)\cdots(n-k+1)$ in the numerator behaves like n^k (each of the k factors is close to n when n is huge and k is fixed), so $\binom{n}{k}p^k \approx \dfrac{n^k}{k!} \cdot \left(\dfrac{\lambda}{n}\right)^k = \dfrac{\lambda^k}{k!}$. Meanwhile, $(1-p)^{n-k} = \left(1 - \dfrac{\lambda}{n}\right)^{n-k}$, and as $n \to \infty$ this converges to $e^{-\lambda}$ (this is exactly the standard limit definition of the exponential function, $\lim_{n\to\infty}(1+x/n)^n = e^x$, applied with $x = -\lambda$, with the small correction of the $-k$ in the exponent vanishing in the limit). Multiplying these two limiting pieces together gives $\dfrac{\lambda^k}{k!}e^{-\lambda}$, the Poisson PMF.

11.5 Worked Example: The Birthday Triple Problem

Worked Example. Approximate probability that three people share a birthday

With n people, each with a birthday uniformly and independently distributed among 365 days (as in the birthday problem from Chapter 3), find the approximate probability that *some* group of three people all share the same birthday.

Setting up via the Poisson Paradigm. For every possible *triplet* of people (there are $\binom{n}{3}$ such triplets), define an event "this triplet all has the same birthday." For a single fixed triplet, the probability that all three share a specific common birthday day is found as follows: fix the first person's birthday as whatever it is; the second person matches it with probability $1/365$; the third person also matches it with probability $1/365$ (independently). So the probability that this specific triplet has a triple birthday match is $(1/365)^2 = 1/365^2$.

Applying the Poisson Paradigm. There are $\binom{n}{3}$ triplets, each with a small probability $1/365^2$ of being a triple match, and these events are only weakly dependent on each other (knowing one triplet matches gives only slight information about another overlapping triplet). So by the Poisson Paradigm, the total number of "triple-match" triplets, call it X, is approximately $\text{Pois}(\lambda)$ where

$$\lambda = \binom{n}{3} \cdot \frac{1}{365^2}.$$

Computing the final probability. We want $\mathbb{P}(\text{at least one triple match}) = \mathbb{P}(X \geq 1)$. Using the complement trick (much easier than summing $\mathbb{P}(X = 1) + \mathbb{P}(X = 2) + \cdots$ directly):

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - e^{-\lambda}.$$

This gives a clean, explicit approximation formula in terms of n alone (via $\lambda = \binom{n}{3}/365^2$), avoiding any need for an exact (and far more complicated) combinatorial calculation.

12. Continuous Distributions and the Uniform Distribution

12.1 From Discrete to Continuous

Every random variable studied so far (Bernoulli, Binomial, Hypergeometric, Geometric, Negative Binomial, Poisson) is **discrete**: it can only take values from a countable list (like the non-negative integers). But many real quantities — a person's exact height, the precise time until an event happens, a measurement error — can in principle take *any* real number in some range, with no gaps. These are **continuous** random variables, and they require new tools, summarized in the comparison table below.

	Discrete	**Continuous**
PMF/PDF	$\mathbb{P}(X = x)$	PDF $f_X(x)$, and $\mathbb{P}(X = x) = 0$
CDF	$F(x) = \mathbb{P}(X \leq x)$	$F_X(x) = \mathbb{P}(X \leq x)$
Expectation	$\mathbb{E}(X) = \sum_x x\,\mathbb{P}(x)$	$\mathbb{E}(X) = \int x f_X(x)\,dx$

The single most important conceptual difference is highlighted in the table: for a continuous random variable, $\mathbb{P}(X = x) = 0$ for *every* individual value x. This might seem strange at first — surely X has to take *some* value? The resolution is that probability for continuous variables is not concentrated at points, but spread smoothly over intervals; asking for the probability of one exact, infinitely-precise real number is like asking for the length of a single point on a ruler — it's zero, even though the ruler as a whole has positive length. We replace the PMF (which directly gave $\mathbb{P}(X = x)$) with a **probability density function** (PDF), which gives probabilities only after being *integrated* over a range.

12.2 The Probability Density Function (PDF)

Definition. PDF

A continuous random variable X has **probability density function** $f(x)$ if, for all real numbers $a \leq b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx.$$

Validity conditions. For f to be a legitimate PDF, we need:

1. $f(x) \geq 0$ for all x (probabilities/densities can't be negative);
2. $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$ (the total probability over all possible values must be 1, matching axiom 1 of probability).

It's worth emphasizing: $f(x)$ itself is *not* a probability (it can even exceed 1 in value, for instance if the distribution is heavily concentrated on a very short interval) — it's a **density**, and only becomes an actual probability after being integrated over some range of x-values.

12.2.1 Relating the PDF and CDF

If X has PDF f, then its CDF is obtained by integrating the density from $-\infty$ up to x:

$$F_X(x) = \int_{-\infty}^{x} f(t)\,dt.$$

Conversely, if X is continuous and we know its CDF F, we can recover the PDF by differentiating:

$$f(x) = F'(x).$$

This is a direct consequence of the Fundamental Theorem of Calculus, which also gives us the useful identity

$$\mathbb{P}(a < X \leq b) = F(b) - F(a)$$

(the probability of landing in an interval is the difference of the CDF at the two endpoints — this follows since $F(b) - F(a) = \int_{-\infty}^{b} f - \int_{-\infty}^{a} f = \int_a^b f$, exactly matching the definition of the PDF's integral over $[a, b]$).

12.3 Variance

Before introducing our first continuous distribution, we define another key summary statistic: **variance**, which measures how spread out a distribution is around its mean.

Definition. Variance and Standard Deviation

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}(X))^2\big] = \mathbb{E}(X^2) - \big(\mathbb{E}(X)\big)^2$$

and $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$.

The first form, $\mathbb{E}[(X - \mathbb{E}(X))^2]$, is the most intuitive definition: it's the average squared distance of X from its own mean (we square the deviation rather than just averaging the deviation itself, because the deviations $X - \mathbb{E}(X)$ average out to exactly zero by definition of the mean, telling us nothing about spread; squaring makes all deviations positive so they don't cancel out, and also penalizes large deviations more heavily). The second form, $\mathbb{E}(X^2) - (\mathbb{E}(X))^2$, is algebraically equivalent (expand $(X - \mathbb{E}(X))^2 = X^2 - 2X\mathbb{E}(X) + (\mathbb{E}(X))^2$, then take expectations term by term using linearity, noting $\mathbb{E}(X)$ is a constant so $\mathbb{E}(2X\mathbb{E}(X)) = 2\mathbb{E}(X)\mathbb{E}(X) = 2(\mathbb{E}(X))^2$; combining gives $\mathbb{E}(X^2) - 2(\mathbb{E}(X))^2 + (\mathbb{E}(X))^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$), and is usually much easier to compute in practice, since it only requires knowing the two simpler quantities $\mathbb{E}(X)$ and $\mathbb{E}(X^2)$.

12.4 The Uniform Distribution

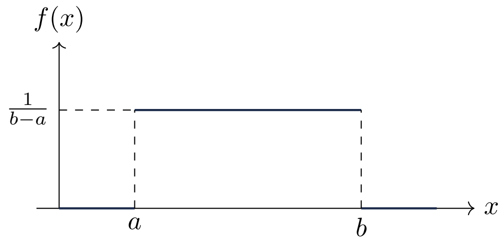
Definition. Uniform Distribution

$\mathrm{Unif}(a, b)$ describes a "completely random" point in the interval $[a, b]$ — meaning every sub-interval of a given length within $[a, b]$ is equally likely to contain the point, regardless of where that sub-interval is located.

12.4.1 The PDF

Since every part of the interval $[a, b]$ should be "equally weighted," the density must be *constant* over $[a, b]$ (and zero outside it). Since the total area under the PDF must equal 1 (validity condition 2), and the width of the interval is $b - a$, the constant height of the density must be $1/(b - a)$ (height times width = area = 1):

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise.} \end{cases}$$



12.4.2 The CDF

Integrating the constant density from a up to x (for $a \le x \le b$):

$$F(x) = \begin{cases} 0, & x < a \\ \dfrac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b. \end{cases}$$

The middle piece comes from $F(x) = \int_a^x \frac{1}{b-a}\,dt = \frac{1}{b-a}(x - a) = \frac{x-a}{b-a}$, which rises linearly from 0 at $x = a$ to 1 at $x = b$, matching the geometric picture of a uniformly rising cumulative probability across the interval.

12.4.3 Expectation

By symmetry (the distribution is perfectly balanced around the midpoint of $[a, b]$), the mean is simply the midpoint:

$$\mathbb{E}(X) = \frac{a+b}{2}.$$

12.5 LOTUS: The Law of the Unconscious Statistician

We have only defined $\mathbb{E}(X)$ directly, but very often we want the expectation of some *function* of X, such as $\mathbb{E}(X^2)$ (needed for variance) or $\mathbb{E}(2^X)$ (as seen in the St. Petersburg paradox). Computing this in general requires knowing the distribution of $g(X)$ itself, which can be complicated to derive. The following theorem provides a shortcut.

For any function g,

$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx.$$

The name is a bit tongue-in-cheek: it says you can compute $\mathbb{E}(g(X))$ by just plugging $g(x)$ into the *original* density of X and integrating, "unconsciously" (i.e., without bothering to first work out the distribution of the new random variable $g(X)$ itself). This theorem (stated here for the continuous case; an analogous discrete version replaces the integral with a sum over x, weighting by $\mathbb{P}(X = x)$) is used constantly throughout the rest of this book.

12.6 Worked Example: Moments of a Uniform(0,1) Random Variable

Worked Example. $U \sim \mathrm{Unif}(0,1)$**: computing** $\mathbb{E}(U)$, $\mathbb{E}(U^2)$**, and** $\mathrm{Var}(U)$

For $U \sim \mathrm{Unif}(0,1)$ (the special case $a = 0, b = 1$), the density is $f(u) = 1$ for $0 \le u \le 1$.

Mean. By the symmetry formula, $\mathbb{E}(U) = \dfrac{0+1}{2} = \dfrac{1}{2}$. (Double-checking via direct integration: $\mathbb{E}(U) = \int_0^1 u \cdot 1\, du = \left[\dfrac{u^2}{2}\right]_0^1 = \dfrac{1}{2}$. Matches.)

Second moment, via LOTUS. Using LOTUS with $g(u) = u^2$:

$$\mathbb{E}(U^2) = \int_0^1 u^2 \cdot 1\, du = \left[\frac{u^3}{3}\right]_0^1 = \frac{1}{3}.$$

Variance. Using the computational formula for variance,

$$\mathrm{Var}(U) = \mathbb{E}(U^2) - \big(\mathbb{E}(U)\big)^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{4-3}{12} = \frac{1}{12}.$$

13. The Normal (Gaussian) Distribution

13.1 Why the Normal Distribution Matters: The Central Limit Theorem

The sum (or average) of a large number of independent, identically distributed random variables — regardless of the shape of their individual distribution — looks approximately like a Normal (Gaussian) distribution.

This remarkable universality is why the Normal distribution shows up everywhere in nature and statistics: measurement errors, heights of people, test scores, and countless other quantities arise as the aggregate (sum) of many small, roughly independent contributing factors, and the Central Limit Theorem tells us that such aggregates tend toward the same bell-shaped curve no matter what the individual contributing factors look like. A full proof of the Central Limit Theorem requires moment generating functions (introduced in Chapter 14), so here we focus on building up the Normal distribution itself.

13.2 The Standard Normal Distribution

Definition. Standard Normal

$Z \sim N(0,1)$ has PDF of the form $f(z) = c\,e^{-z^2/2}$ for some normalizing constant c (chosen to make the total area equal to 1).

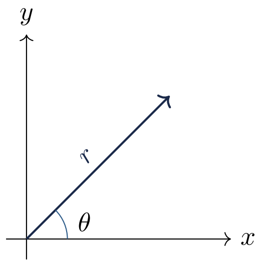
13.2.1 Finding the Normalizing Constant

We need $\displaystyle\int_{-\infty}^{\infty} e^{-z^2/2}\,dz = \frac{1}{c}$. This integral cannot be evaluated by ordinary single-variable techniques (there's no elementary antiderivative for $e^{-z^2/2}$) — instead, we use a classic and beautiful trick: square the integral, turning it into a two-dimensional integral, and then switch to polar coordinates.

Step 1: Square the integral. Let $I = \displaystyle\int_{-\infty}^{\infty} e^{-z^2/2}\,dz$. Then

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2/2}dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}dy\right) \\ &= \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\,dx\,dy, \end{aligned}$$

where we renamed the dummy integration variable in the second factor to y (it's still the same integral, just relabeled), and combined the two separate one-dimensional integrals into a single two-dimensional integral over the entire xy-plane, using the fact that $e^{-x^2/2}e^{-y^2/2} = e^{-(x^2+y^2)/2}$.



Step 2: Convert to polar coordinates. Using $x = r\cos\theta$, $y = r\sin\theta$, so $x^2 + y^2 = r^2$, and the area element transforms as $dx\,dy = r\,dr\,d\theta$ (the extra factor of r is the standard Jacobian for polar coordinates, accounting for the fact that a small "polar rectangle" of angular width $d\theta$ and radial width dr has area $r\,dr\,d\theta$, growing with distance from the origin), with r ranging over $[0,\infty)$ and θ ranging over $[0,2\pi)$ to cover the whole plane:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2}\,r\,dr\,d\theta.$$

Step 3: Evaluate the radial integral via substitution. Let $u = r^2/2$, so $du = r\,dr$ (exactly matching the $r\,dr$ already present in the integral). As r ranges from 0 to ∞, u also ranges from 0 to ∞. So

$$\int_0^{\infty} e^{-r^2/2} r\,dr = \int_0^{\infty} e^{-u}\,du = \big[-e^{-u}\big]_0^{\infty} = 0 - (-1) = 1.$$

Step 4: Evaluate the angular integral.

$$I^2 = \int_0^{2\pi} 1\,d\theta = 2\pi.$$

Step 5: Solve for I and then c. Since $I^2 = 2\pi$ and $I = \int e^{-z^2/2} dz$ is manifestly positive (the integrand is always positive), $I = \sqrt{2\pi}$. Since $I = 1/c$, we get $c = 1/\sqrt{2\pi}$. So the standard Normal PDF is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

13.2.2 Mean of the Standard Normal

By symmetry: $f(z)$ is an **even function** (i.e., $f(-z) = f(z)$, since $(-z)^2 = z^2$), so the function $z \cdot f(z)$ appearing inside the integral for $\mathbb{E}(Z)$ is an **odd function** (negating z flips its sign: $(-z)f(-z) = -zf(z)$). The integral of any odd function over a symmetric interval like $(-\infty, \infty)$ is exactly 0 (the positive and negative halves cancel perfectly). So

$$\mathbb{E}(Z) = \int_{-\infty}^{\infty} z f(z)\, dz = 0.$$

13.2.3 Variance of the Standard Normal

We compute $\mathbb{E}(Z^2)$ using **integration by parts**, the standard calculus technique for integrating a product of two functions, $\int u\, dv = uv - \int v\, du$.

$$\mathbb{E}(Z^2) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz.$$

Set $u = z$ (so $du = dz$) and $dv = ze^{-z^2/2}\, dz$. To find v, we need an antiderivative of $ze^{-z^2/2}$: notice that $\frac{d}{dz}\left[-e^{-z^2/2}\right] = -e^{-z^2/2} \cdot (-z) = ze^{-z^2/2}$ (using the chain rule), so $v = -e^{-z^2/2}$. Applying integration by parts:

$$\int_{-\infty}^{\infty} z \cdot ze^{-z^2/2}\, dz = \Big[-ze^{-z^2/2}\Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \big(-e^{-z^2/2}\big)\, dz.$$

The boundary term $\left[-ze^{-z^2/2}\right]_{-\infty}^{\infty}$ vanishes: as $z \to \pm\infty$, the exponential $e^{-z^2/2}$ decays to 0 much faster than the linear factor z grows, so the product goes to 0 at both ends. This leaves

$$\int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz = \int_{-\infty}^{\infty} e^{-z^2/2}\, dz = \sqrt{2\pi}$$

(using the integral we computed earlier in this chapter). So

$$\mathbb{E}(Z^2) = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1.$$

Since $\mathbb{E}(Z) = 0$, the variance is

$$\mathrm{Var}(Z) = \mathbb{E}(Z^2) - (\mathbb{E}(Z))^2 = 1 - 0 = 1.$$

This confirms the notation $N(0, 1)$: mean 0, variance 1.

13.3 Notation: The Standard Normal CDF

Because the standard Normal PDF has no elementary antiderivative, its CDF is given a special name and symbol rather than a closed formula:

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-t^2/2}\, dt.$$

Values of $\Phi(z)$ are typically looked up in tables or computed numerically, rather than via an explicit formula.

13.4 Location and Scale: The General Normal Distribution

We now build the general Normal distribution $N(\mu, \sigma^2)$ by transforming the standard Normal.

Definition. General Normal

Let $X = \mu + \sigma Z$, where $Z \sim N(0, 1)$, $\mu \in \mathbb{R}$ (called the **location** or **mean** parameter), and $\sigma > 0$ (called the **scale** or **standard deviation** parameter). Then $X \sim N(\mu, \sigma^2)$.

13.4.1 Mean and Variance of X

To find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$, we need two general facts about how variance behaves under shifting and scaling, which we state and justify here:

- $\mathrm{Var}(X + c) = \mathrm{Var}(X)$ for any constant c: adding a constant shifts every value of X by the same amount, which shifts the mean by exactly c too, so the *deviations from the mean*, $X - \mathbb{E}(X)$, are completely unchanged; hence the variance (which only depends on these deviations) is unchanged.
- $\mathrm{Var}(cX) = c^2\mathrm{Var}(X)$ for any constant c: scaling X by c scales every deviation from the mean by c as well, and since variance involves *squared* deviations, the scaling factor gets squared too.

- $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ if X, Y are independent: this is analogous to linearity of expectation, but requires independence (unlike linearity of expectation, which needs no such assumption) — this fact is proven properly once covariance is introduced, but we use it here as a standard tool.

Applying linearity of expectation to $X = \mu + \sigma Z$:

$$\mathbb{E}(X) = \mathbb{E}(\mu) + \sigma\mathbb{E}(Z) = \mu + \sigma \cdot 0 = \mu$$

(the expectation of a constant μ is just μ itself, since it doesn't vary). And for variance, using $\mathrm{Var}(X+c) = \mathrm{Var}(X)$ and $\mathrm{Var}(cX) = c^2\mathrm{Var}(X)$:

$$\mathrm{Var}(X) = \mathrm{Var}(\mu + \sigma Z) = \mathrm{Var}(\sigma Z) = \sigma^2\mathrm{Var}(Z) = \sigma^2 \cdot 1 = \sigma^2.$$

This confirms that μ truly is the mean and σ^2 truly is the variance of $X \sim N(\mu, \sigma^2)$, justifying the notation.

13.4.2 Deriving the General Normal PDF

To find the PDF of $X \sim N(\mu, \sigma^2)$, we use the technique of **standardization**: express X's CDF in terms of the already-known standard Normal CDF Φ, then differentiate.

Since $X = \mu + \sigma Z$, we can solve for Z: $Z = \dfrac{X-\mu}{\sigma}$. Then

$$F_X(x) = \mathbb{P}(X \le x) = \mathbb{P}\left(Z \le \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right),$$

where the middle step is simple algebra (subtracting μ and dividing by the positive constant σ preserves the direction of the inequality), and the last step is just the definition of Φ as the CDF of a standard Normal, evaluated at the point $\frac{x-\mu}{\sigma}$.

To get the PDF, differentiate with respect to x, using the chain rule (since the argument of Φ, namely $\frac{x-\mu}{\sigma}$, itself depends on x):

$$f_X(x) = \frac{d}{dx}\Phi\left(\frac{x-\mu}{\sigma}\right) = \Phi'\left(\frac{x-\mu}{\sigma}\right) \cdot \frac{1}{\sigma}.$$

Since $\Phi' = f$ (the standard Normal PDF, by the PDF–CDF relationship from Chapter 12), and $f(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$:

$$\begin{aligned} f_X(x) &= \frac{1}{\sigma} \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \\ &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \end{aligned}$$

This is the famous "bell curve" formula. The parameter μ shifts the center of the bell horizontally, and σ controls how wide or narrow the bell is (larger σ means a wider, flatter, more spread-out curve, since the data is more variable).

14. Variance Revisited, and the Exponential Distribution

14.1 Variance of the Poisson Distribution

If $X \sim \text{Pois}(\lambda)$, then $\text{Var}(X) = \lambda$.

Step 1: Find $\mathbb{E}(X^2)$. A useful trick for Poisson-type sums is to first compute $\mathbb{E}[X(X-1)]$ (rather than $\mathbb{E}(X^2)$ directly), because the factor $X(X-1)$ interacts nicely with the factorial in the denominator of the Poisson PMF:

$$\mathbb{E}[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\frac{e^{-\lambda}\lambda^k}{k!}.$$

The $k = 0$ and $k = 1$ terms both vanish (multiplied by $k(k-1) = 0$ in each case), so we sum from $k = 2$. For $k \geq 2$, $\dfrac{k(k-1)}{k!} = \dfrac{1}{(k-2)!}$ (since $k! = k(k-1)(k-2)!$). So

$$\mathbb{E}[X(X-1)] = e^{-\lambda}\lambda^2 \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = e^{-\lambda}\lambda^2 e^{\lambda} = \lambda^2,$$

re-indexing with $j = k-2$ and again using the exponential Taylor series.

Step 2: Relate $\mathbb{E}[X(X-1)]$ to $\mathbb{E}(X^2)$. Expanding, $X(X-1) = X^2 - X$, so by linearity of expectation, $\mathbb{E}[X(X-1)] = \mathbb{E}(X^2) - \mathbb{E}(X)$, which rearranges to

$$\mathbb{E}(X^2) = \mathbb{E}[X(X-1)] + \mathbb{E}(X) = \lambda^2 + \lambda.$$

Step 3: Compute the variance.

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda.$$

A striking and easy-to-remember fact about the Poisson distribution: its mean and its variance are both equal to λ.

14.2 Variance of the Binomial Distribution

If $X \sim \text{Bin}(n, p)$, then $\text{Var}(X) = np(1-p) = npq$.

Setup using indicators. As before, write $X = I_1 + I_2 + \cdots + I_n$ where the I_i are i.i.d. $\text{Bern}(p)$. Square this sum:

$$X^2 = \left(\sum_i I_i\right)^2 = \sum_i I_i^2 + 2\sum_{i<j} I_i I_j,$$

expanding the square of a sum in the usual way: every term is either a "diagonal" term $I_i \cdot I_i = I_i^2$, or an "off-diagonal" cross term $I_i I_j$ (with $i \neq j$), and each unordered pair $\{i, j\}$ with $i < j$ appears exactly twice in the full expansion (once as $I_i I_j$ and once as $I_j I_i$), which is why the cross-term sum carries a factor of 2.

Simplifying I_i^2. Since I_i only takes values 0 or 1, $I_i^2 = I_i$ (squaring 0 gives 0, squaring 1 gives 1 — the value is unchanged). So $\mathbb{E}(I_i^2) = \mathbb{E}(I_i) = p$.

Simplifying $\mathbb{E}(I_i I_j)$ for $i \neq j$. Since the I_i are independent, $I_i I_j = 1$ exactly when both $I_i = 1$ and $I_j = 1$, which (by independence) has probability $p \cdot p = p^2$. So $\mathbb{E}(I_i I_j) = p^2$ (this also follows from the general fact that for independent random variables, $\mathbb{E}(I_i I_j) = \mathbb{E}(I_i)\mathbb{E}(I_j)$).

Putting it together. There are n diagonal terms (each contributing p in expectation) and $\binom{n}{2} = \frac{n(n-1)}{2}$ off-diagonal unordered pairs, each appearing with coefficient 2 in the original expansion and contributing p^2 in expectation, so the total cross-term contribution is $2\binom{n}{2}p^2 = n(n-1)p^2$. So

$$\mathbb{E}(X^2) = np + n(n-1)p^2.$$

Therefore

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \\ &= \left[np + n(n-1)p^2\right] - (np)^2 \\ &= np + n^2p^2 - np^2 - n^2p^2 \\ &= np - np^2 = np(1-p) = npq. \end{aligned}$$

(The n^2p^2 terms cancel exactly.) Notice the variance is maximized when $p = 1/2$ (where $pq = 1/4$ is largest) and shrinks to 0 as $p \to 0$ or $p \to 1$ (where the outcome becomes nearly deterministic — almost always failure, or almost always success, leaving little room for variability).

14.3 Building the Exponential Distribution from the Poisson Process

Worked Example. Time until the first email arrives

Suppose emails arrive over time according to a process where the number of emails received in a time window of length t is $N_t \sim \text{Pois}(\lambda t)$ (a natural extension of the Poisson distribution to a continuous time axis, where λ is the average arrival rate per unit time). Let T be the (continuous) random variable giving the time until the very first email arrives. We want the distribution of T.

Key observation. The event "$T > t$" (no email has arrived yet by time t) is exactly the same as the event "$N_t = 0$" (zero emails arrived in the window $[0, t]$). So

$$\mathbb{P}(T > t) = \mathbb{P}(N_t = 0) = \frac{e^{-\lambda t}(\lambda t)^0}{0!} = e^{-\lambda t},$$

plugging $k = 0$ into the Poisson PMF with rate parameter λt (since the number of arrivals in a window of length t has Poisson parameter λt, not just λ, because the rate λ accumulates over the duration t).

Finding the CDF and PDF. Since $F(t) = \mathbb{P}(T \le t) = 1 - \mathbb{P}(T > t)$,

$$F(t) = 1 - e^{-\lambda t}, \qquad t \ge 0.$$

Differentiating to get the PDF:

$$f(t) = F'(t) = \lambda e^{-\lambda t}, \qquad t \ge 0.$$

This derivation shows precisely how the continuous **Exponential distribution** arises naturally as "the waiting time until the first event" in a Poisson-type arrival process — a continuous-time analogue of how the discrete Geometric distribution arose as "the number of trials until the first success."

14.4 The Exponential Distribution

Definition. Exponential Distribution

$X \sim \text{Expo}(\lambda)$ has PDF and CDF

$$f(x) = \lambda e^{-\lambda x}, \qquad F(x) = 1 - e^{-\lambda x}, \qquad x > 0.$$

14.4.1 The Scaling Property

If $X \sim \text{Expo}(\lambda)$, then $Y = \lambda X \sim \text{Expo}(1)$.

Why: $\mathbb{P}(Y \le y) = \mathbb{P}(\lambda X \le y) = \mathbb{P}(X \le y/\lambda) = 1 - e^{-\lambda(y/\lambda)} = 1 - e^{-y}$, which is exactly the Expo(1) CDF. This property is extremely convenient: it means we only need to work out the moments of the simplest case Expo(1), and then rescale to get the moments of any Expo(λ).

14.4.2 Moments of Expo(1), and Then General Expo(λ)

For $Y \sim \text{Expo}(1)$ (density e^{-y}), one can show via integration by parts (similar in spirit to the Normal-distribution calculations in Chapter 13, though we state the results directly here since the techniques are by now familiar) that

$$\mathbb{E}(Y) = 1, \qquad \mathbb{E}(Y^2) = 2,$$

$$\text{Var}(Y) = \mathbb{E}(Y^2) - (\mathbb{E}(Y))^2 = 2 - 1 = 1.$$

Since $X = Y/\lambda$ (inverting the scaling relationship $Y = \lambda X$), we use linearity of expectation and the scaling rule for variance:

$$\mathbb{E}(X) = \frac{1}{\lambda}\mathbb{E}(Y) = \frac{1}{\lambda}, \qquad \text{Var}(X) = \frac{1}{\lambda^2}\text{Var}(Y) = \frac{1}{\lambda^2}.$$

These are important and frequently-used formulas: the mean waiting time for an Expo(λ) random variable is $1/\lambda$ (matching the intuitive idea that a higher rate λ means a shorter expected wait), and the variance is $1/\lambda^2$.

14.5 The Memoryless Property

This is perhaps the single most distinctive and important feature of the Exponential distribution.

If $X \sim \text{Expo}(\lambda)$, then for all $s, t \ge 0$,

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

Proof. By the definition of conditional probability,

$$\mathbb{P}(X > s + t \mid X > s) = \frac{\mathbb{P}(X > s + t \text{ and } X > s)}{\mathbb{P}(X > s)} = \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)},$$

where the numerator simplifies because the event "$X > s + t$" already implies "$X > s$" (since $s + t \ge s$), so their intersection is just "$X > s+t$" itself. Now, since $\mathbb{P}(X > x) = 1 - F(x) = 1 - (1 - e^{-\lambda x}) = e^{-\lambda x}$,

$$\frac{\mathbb{P}(X > s+t)}{\mathbb{P}(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda(s+t)+\lambda s} = e^{-\lambda t} = \mathbb{P}(X > t). \qquad \blacksquare$$

Interpretation. This says that if X represents a waiting time (say, the lifetime of a lightbulb, or the time until the next email arrives), and we already know the wait has lasted at least s units of time without the event occurring, then the *additional* waiting time needed (measured from now) has exactly the same distribution as if we were starting fresh from time zero — the process has no "memory" of how long it has already been waiting. This is a very special and somewhat counterintuitive property: it implies, for instance, that a lightbulb modeled as Exponential does not "wear out" or become more likely to fail the longer it has already lasted — its remaining lifespan always looks the same, regardless of its age so far. (The Exponential distribution is, in fact, the *only* continuous distribution with this memoryless property.)

14.5.1 Corollary: Expected Remaining Wait Given Survival

A direct and useful consequence of the memoryless property:

If $X \sim \text{Expo}(\lambda)$, then for any $a \geq 0$,

$$\mathbb{E}(X \mid X > a) = a + \frac{1}{\lambda}.$$

This follows immediately from the memoryless property: conditional on having survived past time a, the *additional* time until X occurs behaves exactly like a fresh $\text{Expo}(\lambda)$ random variable (with mean $1/\lambda$, as derived above), so the total time (measured from the start) is a (the time already elapsed) plus this fresh expectation $1/\lambda$.

14.6 Looking Ahead: Moment Generating Functions

We have now computed the mean and variance of several distributions, each time using a somewhat different (and sometimes quite involved) ad hoc trick: summing series for the discrete distributions, integrating by parts for the Normal and Exponential distributions, conditioning arguments for the Geometric distribution, and so on. The next chapter introduces a single unifying tool — the **moment generating function** — that can systematically produce *all* of the moments ($\mathbb{E}(X)$, $\mathbb{E}(X^2)$, $\mathbb{E}(X^3)$, and so on) of a distribution at once, and which also gives us a powerful method for proving that the sum of independent random variables follows a particular distribution, generalizing the convolution-based proof we saw for the Binomial distribution in Chapter 8.

15. Moment Generating Functions

15.1 Definition

Definition. Moment Generating Function

A random variable X has **moment generating function** (MGF)

$$M(t) = \mathbb{E}(e^{tX}),$$

viewed as a function of the real variable t, provided this expectation is finite for t in some open interval $(-a, a)$ around 0, for some $a > 0$.

The MGF is a single function that, as we will see, packages up *all* of the moments of X $(\mathbb{E}(X), \mathbb{E}(X^2), \mathbb{E}(X^3), \ldots)$ simultaneously, and has remarkably convenient algebraic properties when working with sums of independent random variables.

15.2 Why Study the MGF? Three Key Reasons

To understand where the MGF's usefulness comes from, expand e^{tX} using its Taylor series, $e^{tX} = \sum_{n=0}^{\infty} \frac{(tX)^n}{n!} = \sum_{n=0}^{\infty} \frac{X^n t^n}{n!}$, and take expectations term by term (using linearity of expectation, applied to the infinite sum):

$$M(t) = \mathbb{E}(e^{tX}) = \sum_{n=0}^{\infty} \frac{\mathbb{E}(X^n)}{n!} t^n.$$

15.2.1 Reason 1: Moments Are Encoded as Taylor Coefficients

Looking at the expansion above, the coefficient of t^n in the Taylor series of $M(t)$ is exactly $\frac{\mathbb{E}(X^n)}{n!}$. Equivalently, recalling that the n-th derivative of a function evaluated at 0, divided by $n!$, gives the coefficient of t^n in its Taylor series (this is just Taylor's theorem from calculus), we get

$$M^{(n)}(0) = \mathbb{E}(X^n).$$

This means: to find any moment of X, just differentiate the MGF n times and plug in $t = 0$. This often turns a hard summation or integration problem (computing $\mathbb{E}(X^n)$ directly from the definition) into an easier differentiation problem.

15.2.2 Reason 2: The MGF Determines the Distribution

A deep (and not at all obvious) fact, which we state without proof here, is that if two random variables X and Y have the *same* MGF (on some common interval around 0), then they have the *same* distribution (same PDF/PMF, same CDF, everything). This means the MGF acts as a "fingerprint": if we can show two random variables have matching MGFs, we know for certain they have the same distribution, without needing to directly compare their PMFs or PDFs. This is exactly the strategy used below to show sums of independent Poissons are Poisson.

15.2.3 Reason 3: MGFs of Sums of Independent Variables Multiply

If X and Y are independent, then

$$M_{X+Y}(t) = M_X(t)\, M_Y(t).$$

Proof. By definition, $M_{X+Y}(t) = \mathbb{E}\big(e^{t(X+Y)}\big) = \mathbb{E}\big(e^{tX} e^{tY}\big)$. Since X and Y are independent, any function of X alone is independent of any function of Y alone (this is a general fact about independence), so e^{tX} and e^{tY} are independent random variables, and the expectation of a product of independent random variables equals the product of their individual expectations:

$$\mathbb{E}\big(e^{tX} e^{tY}\big) = \mathbb{E}(e^{tX})\, \mathbb{E}(e^{tY}) = M_X(t)\, M_Y(t). \qquad \blacksquare$$

This turns the (potentially complicated) convolution-sum computation we performed by hand in Chapter 8 (to show Binomials add) into simple *multiplication of two functions* — a vastly simpler operation, as the example below with Poissons demonstrates.

15.3 MGFs of Familiar Distributions

We list, without full derivation (these follow from direct computation of $\mathbb{E}(e^{tX})$ using each distribution's PMF or PDF, analogous to computing any other expectation via the definition or LOTUS), the MGFs of the distributions covered so far:

Distribution	**MGF** $M(t)$
$X \sim \text{Bern}(p)$	$q + pe^t$
$X \sim \text{Bin}(n, p)$	$(pe^t + q)^n$
$Z \sim N(0, 1)$	$e^{t^2/2}$
$X \sim \text{Expo}(\lambda)$	$\dfrac{\lambda}{\lambda - t}$, for $t < \lambda$
$X \sim \text{Pois}(\lambda)$	$e^{\lambda(e^t - 1)}$

It's worth briefly noting the structural connection between the Bernoulli and Binomial MGFs: since $\text{Bin}(n, p)$ is a sum of n i.i.d. $\text{Bern}(p)$ random variables, by the MGF-of-a-sum property, $M_{\text{Bin}(n,p)}(t)$ should equal $\big(M_{\text{Bern}(p)}(t)\big)^n = (q+pe^t)^n$ — and indeed, this exactly matches the Binomial MGF listed above, providing a nice consistency check on both formulas (and another, even quicker, proof that the sum of n i.i.d. Bernoullis is Binomial, alongside the story-proof and convolution proofs already given).

15.4 Worked Example: Sum of Independent Poissons

Worked Example. Showing the sum of two independent Poissons is Poisson

Let $X \sim \text{Pois}(\lambda)$ and $Y \sim \text{Pois}(\mu)$ be independent. We use MGFs to find the distribution of $X + Y$.
By the MGF-of-a-sum property,

$$M_{X+Y}(t) = M_X(t)\, M_Y(t) = e^{\lambda(e^t-1)} \cdot e^{\mu(e^t-1)}.$$

Since both factors are exponentials with the same base e, we add their exponents:

$$M_{X+Y}(t) = e^{\lambda(e^t-1)+\mu(e^t-1)} = e^{(\lambda+\mu)(e^t-1)}.$$

But this is *exactly* the MGF of a $\text{Pois}(\lambda + \mu)$ random variable, matching the table above with rate parameter $\lambda + \mu$ in place of λ. By Reason 2 above (the MGF determines the distribution uniquely), we conclude

$$X + Y \sim \text{Pois}(\lambda + \mu).$$

This result makes excellent intuitive sense too: if X counts events from one Poisson process with rate λ and Y counts events from an independent Poisson process with rate μ, then pooling both processes together should give a combined process with the combined rate $\lambda + \mu$ — and the MGF calculation confirms this rigorously, with far less algebraic effort than a direct convolution sum (compare to the Binomial convolution proof in Chapter 8, which required invoking Vandermonde's identity) would have required.

16. Joint, Marginal, and Conditional Distributions

16.1 Motivation: Describing Multiple Random Variables at Once

So far we've studied one random variable at a time. But often we care about *several* random variables defined on the same experiment simultaneously — for instance, a person's height and weight, or the x- and y-coordinates of a randomly thrown dart. To describe how two (or more) random variables behave *together*, including any dependence between them, we need the notion of a **joint distribution**.

16.1.1 Joint CDF and Joint PMF

Definition. Joint CDF

For two random variables X and Y, the **joint cumulative distribution function** is

$$F(x, y) = \mathbb{P}(X \le x,\ Y \le y).$$

Definition. Joint PMF, discrete case

If X and Y are both discrete, their **joint PMF** is simply $\mathbb{P}(X = x, Y = y)$, the probability that X and Y simultaneously take the specific values x and y.

16.1.2 Marginal Distributions

Given the *joint* behavior of X and Y, we can recover the distribution of X *alone* (ignoring Y entirely) by summing or integrating out Y. This recovered individual distribution is called a **marginal distribution**.

Definition. Marginal Distribution

In the discrete case, the marginal PMF of X is obtained by summing the joint PMF over all values of Y:

$$\mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x, Y = y).$$

In the continuous case, the marginal density of X is obtained by integrating the joint density over all values of y:

$$f_X(x) = \int f(x, y)\, dy.$$

The logic behind summing/integrating out Y is simply the Law of Total Probability (Chapter 5) applied with the partition given by all the different possible values of Y: the event "$X = x$" can happen together with Y taking any of its possible values, and these different "$Y = y$" scenarios are mutually exclusive and exhaustive, so we sum the joint probabilities of "$X = x$ and $Y = y$" over all y to get the total (marginal) probability of "$X = x$" alone.

16.1.3 Joint PDF (Continuous Case)

Definition. Joint PDF

For continuous X, Y, the joint PDF $f(x, y)$ satisfies, for any region B in the plane,

$$\mathbb{P}((X, Y) \in B) = \iint_B f(x, y)\, dx\, dy.$$

This is the direct two-dimensional analogue of the single-variable PDF from Chapter 12: instead of integrating over an interval to get the probability of landing in that interval, we integrate over a two-dimensional region B to get the probability that the pair (X, Y) lands somewhere in that region.

16.2 Independence of Random Variables, Revisited

We can now restate the independence condition from Chapter 9 in terms of the joint CDF:

X and Y are independent if and only if $F(x, y) = F_X(x)\, F_Y(y)$ for all x, y.

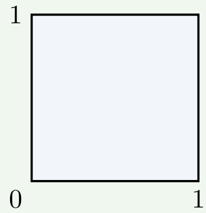
This says the joint CDF factors perfectly into the product of the two marginal CDFs — exactly mirroring the definition of independent events from Chapter 4 ($\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$), but now applied to the events "$X \le x$" and "$Y \le y$" for every possible pair of thresholds (x, y) simultaneously. An equivalent and often more directly checkable condition in the continuous case is that the joint *density* factors as $f(x, y) = f_X(x) f_Y(y)$.

16.3 Worked Example: Uniform Distribution on the Unit Square

Worked Example. Independent uniforms

Let (X, Y) be uniformly distributed on the unit square $\{(x, y) : x, y \in [0, 1]\}$ — meaning every sub-region of the square is "weighted" by its area, with no part of the square more likely than any other part of equal area. Since the total area of the square is $1 \times 1 = 1$, and the joint density must be constant over the square and integrate to 1 over that area, the joint density is simply

$$f(x, y) = \begin{cases} 1, & (x, y) \in [0, 1] \times [0, 1] \\ 0, & \text{otherwise.} \end{cases}$$



Finding the marginal of X. Integrate the joint density over all $y \in [0, 1]$ (the only range where the density is nonzero):

$$f_X(x) = \int_0^1 1\, dy = 1, \qquad 0 \le x \le 1.$$

This is exactly the $\text{Unif}(0, 1)$ density. By an identical computation (integrating over x instead), $Y \sim \text{Unif}(0, 1)$ as well.

Checking independence. Since $f(x, y) = 1 = 1 \cdot 1 = f_X(x)\, f_Y(y)$ for all (x, y) in the square (and $f(x, y) = 0 = f_X(x) f_Y(y)$ outside it, since at least one marginal density is 0 there), the joint density factors perfectly into the product of the marginals. Therefore X and Y *are independent*.

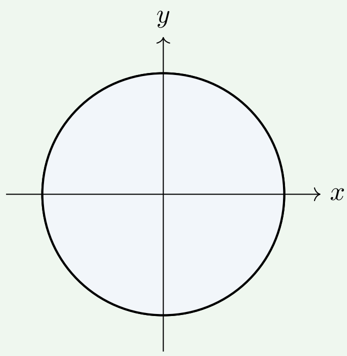
16.4 Worked Example: Uniform Distribution on the Unit Disk

This example is specifically chosen to contrast with the square example: it shows a case where the marginal distributions look deceptively simple, but the variables are *not* independent.

Worked Example. Dependent, despite a simple-looking joint density

Let (X, Y) be uniform on the unit disk $\{(x, y) : x^2 + y^2 \le 1\}$. The area of the unit disk is $\pi \cdot 1^2 = \pi$ (the usual formula for the area of a circle of radius 1), so for the joint density to be constant on the disk and integrate to 1,

$$f(x, y) = \begin{cases} \frac{1}{\pi}, & x^2 + y^2 \le 1 \\ 0, & \text{otherwise.} \end{cases}$$



Finding the marginal of X. For a fixed value of x with $-1 \le x \le 1$, the disk's condition $x^2 + y^2 \le 1$ becomes $y^2 \le 1 - x^2$, i.e., $-\sqrt{1 - x^2} \le y \le \sqrt{1 - x^2}$. So we integrate the constant density $1/\pi$ over exactly this range of y-values, and the length of this range is $2\sqrt{1 - x^2}$ (from $-\sqrt{1 - x^2}$ to $+\sqrt{1 - x^2}$):

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi}\, dy = \frac{2\sqrt{1 - x^2}}{\pi}$$

for $-1 \le x \le 1$. (This is a specific, non-uniform density — it's largest at $x = 0$, where the disk is "tallest" in the y-direction, and tapers to 0 at the edges $x = \pm 1$, where the disk pinches to a single point.)

Checking independence. Compare $f_X(x) f_Y(y)$ (which, by an analogous computation, has Y's marginal density $f_Y(y) = \frac{2\sqrt{1-y^2}}{\pi}$, so the product is $\frac{4\sqrt{1-x^2}\sqrt{1-y^2}}{\pi^2}$) against the actual joint density $f(x, y) = 1/\pi$ on the disk. These are clearly *not* equal in general (the product of marginals varies with both x and y in a complicated way, while the joint density is simply constant on the disk). So

$$f(x, y) \neq f_X(x) f_Y(y),$$

meaning X and Y are **not independent**.
Why does this make intuitive sense? If you're told X is close to 1 (near the right edge of the disk), then Y is forced to be very close to 0 (since the disk is very narrow near its edge) — knowing X gives real information constraining Y. This is the opposite of independence, where knowing one variable should tell you nothing about the other. This contrasts directly with the square example, where knowing X tells you nothing at all about Y (since for *any* value of X in $[0, 1]$, Y can still be anything in $[0, 1]$ with the same uniform density) — the key structural difference is that in the square, the *range* of allowed y-values doesn't depend on x, whereas in the disk, the allowed range of y shrinks as $|x|$ grows.

16.5 Conditional Distributions (Continuous Case)

Just as we defined conditional probability for events, we can define a conditional *distribution* for one continuous random variable given the value of another.

Definition. Conditional PDF

$$f_{Y|X}(y \mid x) = \frac{f(x, y)}{f_X(x)}.$$

This mirrors the definition of conditional probability from Chapter 5 exactly ($\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$), just translated into the language of densities: we take the joint density (the "intersection," in a loose sense) and divide by the marginal density of the variable we're conditioning on.

16.5.1 Worked Example: Conditional Distribution on the Unit Disk

For the uniform-disk example above, fix a particular value of x (with $-1 < x < 1$) and find the conditional density of Y given $X = x$:

$$f_{Y|X}(y \mid x) = \frac{f(x, y)}{f_X(x)} = \frac{1/\pi}{2\sqrt{1 - x^2}/\pi} = \frac{1}{2\sqrt{1 - x^2}}$$

for $-\sqrt{1 - x^2} \leq y \leq \sqrt{1 - x^2}$. This is a *constant* density over the interval $[-\sqrt{1 - x^2}, \sqrt{1 - x^2}]$ (which has length $2\sqrt{1 - x^2}$, exactly matching the reciprocal we found, confirming the density integrates to 1 over this interval). A constant density over an interval is, by definition, a Uniform distribution. So:

$$Y \mid X = x \;\sim\; \text{Unif}\big(-\sqrt{1 - x^2},\ \sqrt{1 - x^2}\big).$$

This is a clean and satisfying geometric result: conditional on knowing the x-coordinate, the y-coordinate is uniformly distributed over exactly the vertical slice of the disk at that x-value — matching our visual intuition of the disk perfectly.

17. Order Statistics and the Multinomial Distribution

17.1 Worked Example: Expected Absolute Difference of Two Uniforms

Worked Example. Computing $\mathbb{E}|X-Y|$ for $X, Y \overset{\text{i.i.d.}}{\sim} \text{Unif}(0,1)$

Let X and Y be independent, each $\text{Unif}(0,1)$. We want $\mathbb{E}|X-Y|$, the expected absolute difference between them.
Setting up via LOTUS (two-variable version). Just as LOTUS (Chapter 12) let us compute $\mathbb{E}(g(X))$ by integrating $g(x)f_X(x)$, the analogous two-variable version lets us compute $\mathbb{E}(g(X,Y))$ by integrating $g(x,y)$ against the joint density $f(x,y)$. Since X, Y are independent uniforms on $[0,1]$, their joint density is $f(x,y) = 1$ on the unit square (as established in Chapter 16), so

$$\mathbb{E}|X-Y| = \iint_{[0,1]^2} |x-y|\,dx\,dy.$$

Splitting the integral by which variable is larger. The absolute value $|x-y|$ is awkward to integrate directly because its formula changes depending on whether $x > y$ or $y > x$. So we split the unit square into two triangular halves: the region where $x > y$ (where $|x-y| = x-y$) and the region where $y > x$ (where $|x-y| = y-x$):

$$\mathbb{E}|X-Y| = \int_0^1 \int_0^x (x-y)\,dy\,dx + \int_0^1 \int_0^y (y-x)\,dx\,dy.$$

(In the first double integral, for a fixed outer value of x, y ranges from 0 to x, covering the triangular region where $y < x$; in the second, the roles are swapped, covering the region where $x < y$.)
Exploiting symmetry. Notice the second integral is identical in structure to the first, just with the roles of x and y swapped (which doesn't change its value, since X and Y have exactly the same distribution, so the picture is symmetric under swapping the two variables). So the two integrals are equal, and we can compute just one and double it:

$$\mathbb{E}|X-Y| = 2\int_0^1 \int_0^x (x-y)\,dy\,dx.$$

Evaluating the inner integral. For fixed x, integrate over y from 0 to x:

$$\int_0^x (x-y)\,dy = \left[xy - \frac{y^2}{2}\right]_0^x = x^2 - \frac{x^2}{2} = \frac{x^2}{2}.$$

Evaluating the outer integral.

$$2\int_0^1 \frac{x^2}{2}\,dx = \int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}.$$

So $\mathbb{E}|X-Y| = \dfrac{1}{3}$.

17.2 Order Statistics: Maximum and Minimum

The calculation above connects beautifully to the idea of **order statistics** — the values of a sample arranged in increasing (or decreasing) order. With just two variables X, Y, the two order statistics are simply the maximum and the minimum.

Definition. Max and Min

Let $M = \max(X,Y)$ and $L = \min(X,Y)$.

Key relationships. There are two simple but extremely useful algebraic identities connecting M, L to X, Y:

- $M - L = |X-Y|$: the gap between the larger and smaller value is exactly the absolute difference — whichever of X, Y happens to be larger, subtracting the smaller from the larger gives a non-negative result equal to $|X-Y|$.
- $M + L = X + Y$: the sum of the max and the min equals the sum of the original two values, since between them, M and L are *literally* just X and Y relabeled in sorted order — no value is created or destroyed by sorting, so the total sum is preserved.

Using these to find $\mathbb{E}(M)$ and $\mathbb{E}(L)$. From the first identity and linearity of expectation, $\mathbb{E}(M) - \mathbb{E}(L) = \mathbb{E}|X-Y| = \frac{1}{3}$ (using our result above). From the second identity, $\mathbb{E}(M) + \mathbb{E}(L) = \mathbb{E}(X) + \mathbb{E}(Y) = \frac{1}{2} + \frac{1}{2} = 1$ (using $\mathbb{E}(X) = \mathbb{E}(Y) = \frac{1}{2}$ for $\text{Unif}(0,1)$, from Chapter 12).
We now have a simple system of two linear equations in two unknowns, $\mathbb{E}(M)$ and $\mathbb{E}(L)$:

$$\mathbb{E}(M) - \mathbb{E}(L) = \frac{1}{3}, \qquad \mathbb{E}(M) + \mathbb{E}(L) = 1.$$

Adding the two equations: $2\mathbb{E}(M) = 1+\frac{1}{3} = \frac{4}{3}$, so $\mathbb{E}(M) = \frac{2}{3}$. Subtracting the first from the second: $2\mathbb{E}(L) = 1-\frac{1}{3} = \frac{2}{3}$, so $\mathbb{E}(L) = \frac{1}{3}$. So:

$$\mathbb{E}(M) = \mathbb{E}(\max(X, Y)) = \frac{2}{3}, \qquad \mathbb{E}(L) = \mathbb{E}(\min(X, Y)) = \frac{1}{3}.$$

This matches intuition nicely: since X, Y are symmetric around $1/2$, the maximum of the two should on average be pulled higher than $1/2$ (to $2/3$) and the minimum pulled lower (to $1/3$), and by the symmetry of the whole setup, these two values are themselves symmetric around $1/2$ (since $\frac{2}{3}$ and $\frac{1}{3}$ average to exactly $\frac{1}{2}$).

17.3 Worked Example: Expected Absolute Difference of Two Standard Normals

Worked Example. Computing $\mathbb{E}|Z_1 - Z_2|$ for $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} N(0,1)$

Step 1: Find the distribution of $Z_1 - Z_2$. Since Z_1 and Z_2 are independent, we can use the fact (a consequence of variance being additive for independent variables, combined with the fact that the difference of two independent Normals is again Normal — a property special to the Normal family related to MGFs from Chapter 15) that $Z_1 - Z_2$ is itself Normally distributed, with mean $\mathbb{E}(Z_1) - \mathbb{E}(Z_2) = 0 - 0 = 0$ and variance $\text{Var}(Z_1) + \text{Var}(-Z_2) = \text{Var}(Z_1) + \text{Var}(Z_2) = 1 + 1 = 2$ (using $\text{Var}(-Z_2) = (-1)^2\text{Var}(Z_2) = \text{Var}(Z_2)$ from the scaling rule for variance in Chapter 13). So

$$Z_1 - Z_2 \sim N(0, 2).$$

Step 2: Re-express in terms of a standard Normal. Since $N(0, 2)$ has variance 2, i.e., standard deviation $\sqrt{2}$, we can write $Z_1 - Z_2 = \sqrt{2}\,Z$ for some $Z \sim N(0, 1)$ (matching the location-scale construction $X = \mu + \sigma Z$ from Chapter 13, here with $\mu = 0, \sigma = \sqrt{2}$).
Step 3: Apply linearity to pull out the constant.

$$\mathbb{E}|Z_1 - Z_2| = \mathbb{E}|\sqrt{2}\,Z| = \sqrt{2}\,\mathbb{E}|Z|$$

(since $\sqrt{2} > 0$, it can be pulled outside the absolute value unchanged, and then outside the expectation as a constant multiplier).
Step 4: Use the known value $\mathbb{E}|Z| = \sqrt{2/\pi}$. This is a standard fact about the standard Normal distribution (derivable via LOTUS, integrating $|z| \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, splitting at $z = 0$ by symmetry, and evaluating the resulting integral; we take this as a known input here rather than re-deriving it, since it requires no new techniques beyond ones already shown). So

$$\mathbb{E}|Z_1 - Z_2| = \sqrt{2} \cdot \sqrt{\frac{2}{\pi}} = \sqrt{\frac{2 \cdot 2}{\pi}} = \sqrt{\frac{4}{\pi}} = \frac{2}{\sqrt{\pi}}.$$

17.4 The Multinomial Distribution

We finish with a generalization of the Binomial distribution to more than two outcome categories.

Definition. Multinomial Distribution — Story

Suppose we perform n independent trials, where each trial falls into exactly one of k categories, with $\mathbb{P}(\text{category } j) = p_j$ for each trial (where $\vec{p} = (p_1, \ldots, p_k)$ is a **probability vector**: each $p_j \geq 0$, and $\sum_j p_j = 1$, since the categories must cover all possibilities for a single trial). Let X_j be the number of the n trials that land in category j. Then the vector $\vec{X} = (X_1, \ldots, X_k)$ has the **Multinomial distribution**, written $\vec{X} \sim \text{Mult}(n, \vec{p})$.

This is a direct generalization of the Binomial story (Chapter 8): the Binomial is the special case $k = 2$ (only two categories, "success" and "failure"), and the Multinomial extends this to any number k of categories.

17.4.1 The Joint PMF

Definition. Multinomial Joint PMF

$$\mathbb{P}(X_1 = n_1, \ldots, X_k = n_k) = \frac{n!}{n_1! \cdots n_k!} p_1^{n_1} \cdots p_k^{n_k}$$

where $n_1 + \cdots + n_k = n$.

Where this comes from. This generalizes the reasoning behind the Binomial PMF (Chapter 8) from two categories to k categories. First, the probability of one *specific* sequence of n trial-outcomes with exactly n_1 trials landing in category 1, n_2 in category 2, and so on, is $p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$ (multiplying the probability of each trial's specific outcome, by independence). Second, we must count how many distinct *orderings* of the n trials produce this same set of category-counts $(n_1, \ldots, n_k)$ — this is a counting problem of arranging n items into k labeled groups of specified sizes $n_1, \ldots, n_k$, which is a direct generalization of the binomial coefficient called a **multinomial coefficient**, equal to $\frac{n!}{n_1! n_2! \cdots n_k!}$ (this formula itself can be derived by first arranging all n trials in some order, $n!$ ways, and then dividing out the $n_1!$ ways of internally re-ordering the (indistinguishable, for counting purposes) trials assigned to category 1 without changing the outcome, the $n_2!$ ways for category 2, and so on, for each category — exactly analogous to how dividing by $k!$ in $\binom{n}{k} = n!/(k!(n-k)!)$ removed the over-counting from internal reordering within the chosen group and the unchosen group). Multiplying the per-sequence probability by the number of equivalent sequences gives the full Multinomial PMF.

17.4.2 Marginal Distributions Are Binomial

For any single category j, the marginal distribution of X_j (ignoring all the other categories) is

$$X_j \sim \mathrm{Bin}(n, p_j), \qquad \mathbb{E}(X_j) = np_j, \qquad \mathrm{Var}(X_j) = np_j(1-p_j).$$

Why this is true. If we only care about whether each trial falls into category j or not (lumping all the *other* $k-1$ categories together into a single "not category j" bucket), then each trial is, from this coarser point of view, simply a Bernoulli trial with "success" probability p_j (landing in category j) and "failure" probability $1 - p_j$ (landing in any other category). Since the n trials are independent, the number of category-j successes among them, X_j, is exactly the number of successes in n independent $\mathrm{Bern}(p_j)$ trials — which is, by definition, $\mathrm{Bin}(n, p_j)$. The mean and variance formulas then follow immediately from the already-established Binomial formulas (Chapters 9 and 14): $\mathbb{E}(X_j) = np_j$ and $\mathrm{Var}(X_j) = np_j(1-p_j)$.

This result is a nice closing illustration of a theme that has run throughout this book: a complicated joint object (the full Multinomial vector $\vec{X}$, describing the simultaneous outcome of all k categories at once) can often be understood by stepping back to a simpler, coarser view (just one category j versus "everything else"), reducing a new and unfamiliar-looking distribution to one we have already fully analyzed.